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A novel Fourier time-sequential PINN approach for multi-frequency analysis of nonlinear aerothermoelastic problems

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ABSTRACT

Panel structures on supersonic vehicles experience severe thermal and aerodynamic loads, leading to nonlinear aerothermoelastic responses that can jeopardize structural safety. Analyzing these responses is particularly challenging because of strong nonlinearity and rich frequency content, especially in long-duration numerical simulations. This work introduces a novel Fourier time-sequential physics-informed neural network (FT-PINN) for aerothermoelastic analysis. By incorporating a primitive function, FT-PINN transforms integro-differential governing equations into partial differential equations. Solution accuracy over long-duration time periods for dynamic analysis is maintained through time-sequential training, normalization, and hard-constraint techniques, while random Fourier feature mapping enhances the model ability to capture complex nonlinear and multi-frequency behaviors. Numerical experiments demonstrate that FT-PINN accurately predicts aerothermoelastic responses under various loading conditions, including thermal buckling, limit cycle oscillations, and both quasi-periodic and chaotic motions. The proposed method reduces relative L_2 error by two to three orders of magnitude compared to existing PINN approaches. It also effectively handles cases involving viscoelastic damping, non-uniform thickness and time-varying parameters, further highlighting its applicability and versatility.

1. Introduction

Multiphysics coupling problems are fundamental in mechanics due to the complex interactions between different physical phenomena, with aerothermoelastic behavior being a persistent challenge. As a classic fluid–structure interaction problem, the aerothermoelastic problem focuses on the stability and nonlinear responses of panels. Dowell [1] was the first to establish a foundational model for two-dimensional panels by considering geometric nonlinearity, using piston theory for aerodynamic effect and a constant in-plane force to represent thermal effect. Considering geometric nonlinearity allows for a more accurate depiction of complex behaviors in highly deformed structures [2].

Subsequently, Wang et al. [3] applied the Galerkin method to analyze aerothermoelastic and damping effects through inter-modal energy transfer. The Rayleigh–Ritz method is also a powerful approach for aerothermoelastic analysis. It has been successfully applied to predict the aerothermoelastic responses of functionally graded plates [4–7], built-up plates [8], as well as metamaterial and orthotropic plates [9–11]. The finite element method is another powerful computational approach for such problems. It has been implemented under both Kirchhoff [12–14] and Mindlin plate theories [15,16], combined with model reduction techniques [17,18]. Across these studies, the stability boundaries of various panels are examined, typically revealing three types of aerothermoelastic responses, i.e., convergence to zero, thermal buckling, and flutter [1,19–21].

Accurately solving the governing equations of aerothermoelastic problems is essential for effectively addressing these challenges. Existing studies typically employ traditional numerical methods, e.g., the Galerkin, Rayleigh–Ritz, and finite element methods. These approaches are well-established and mature, offering reliable accuracy and

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computational efficiency. However, these methods always require cumbersome process for discretizing governing equations and generating meshes. Recently, with the rapid advancement of artificial intelligence, some deep learning methods are proposed for solving governing equations of engineering problems. Of these approaches, physics-informed neural networks (PINNs) have shown considerable potential and are attracting growing research interest [22]. As a meshless technique for solving partial differential equations (PDEs) without the need for equation discretization, PINNs offer a promising computing paradigm for aerothermoelastic analysis.

Basically, PINNs learn the mapping from spatiotemporal coordinates to the solutions of PDEs by incorporating physical laws, such as governing equations and initial and boundary conditions, to constrain the neural network training process. Since the introduction by Raissi et al. [22], it has been gradually improved for better accuracy and efficiency through various techniques including hard constraints [23–25], new network architectures [26,27] and novel training strategies [28,29]. Hitherto, PINN methods have been successfully applied to various fields such as industrial machinery [30–32], civil engineering [33–35], solid mechanics [36,37] and fluid mechanics [38,39]. In addition, PINNs have also demonstrated practical utility in structural dynamics. Yan et al. [40] extended PINNs to analyze the natural vibration characteristics of composite structures by integrating the extreme learning machine. For structural response modeling, excellent performance in linear and nonlinear systems has been achieved by introducing long short-term memory networks and physics constraints into the framework, resulting in the PI-LSTM method [41]. Meanwhile, to establish efficient surrogate models for nonlinear structural dynamics, researchers have combined PINNs with traditional time-stepping schemes. This includes the E-PINN method that incorporates an explicit time-domain method [42], and the RK-PINN method that integrates the Runge–Kutta algorithm with a nerual network framework [43]. Both have been validated for effective dynamic modeling and analysis.

Despite these advancements, existing PINN-based methods are still fraught with several limitations. One significant drawback is error divergence over time, which hinders their ability to provide accurate predictions in long-duration simulations. To address this issue, some domain decomposition techniques and time-sequential training strategies have been proposed [44–49]. These approaches divide the time domain into several short time segments so as to reduce the solving difficulties in each time segment and thus raise the solution accuracy. Spectral bias is another disadvantage of PINNs. This makes the fully-connected PINN to more readily learn low-frequency functions, while struggling to capture high-frequency functions [50,51]. To resolve this problem, Liu et al. [52] proposed the MscaleDNN method by using multiple neural networks. The input data were scaled by different ratios and then passed to different networks. Although this method requires more computational effort, it enhances solution accuracy for dynamic problems with multiple frequencies. In addition, Wang et al. [26] introduced random Fourier feature mapping into PINNs, resulting in the development of two methods (i.e., MFF-PINN and ST-MFF-PINN). They showed that the random Fourier feature mapping technique enhances the neural network's ability to capture multi-frequency responses. Then some similar approaches integrating PINNs with the random Fourier feature mapping technique were developed and applied to structural dynamics [53,54]. However, without known data within the solution domain, these methods are effective only for short-duration simulations.

The aerothermoelastic problem investigated in this work exhibits strong nonlinearity and multi-frequency characteristics, and typically requires long-duration simulations to capture the variation of steady-state responses. Nevertheless, the preceding review highlights that existing PINN methods are unable to provide accurate long-duration solutions for nonlinear problems exhibiting multi-frequency characteristics. Therefore, this study aims to develop an effective approach within the PINN framework for accurate aerothermoelastic analysis. The main novelties of the proposed method are highlighted as follows:

- A new FT-PINN method that integrates a time-sequential training strategy with random Fourier feature mapping is introduced.
- The use of random Fourier feature mapping enhances the model ability to capture the nonlinear and multi-frequency characteristics inherent in aerothermoelastic responses, while the time-sequential training strategy effectively reduces learning complexity for such responses within each time segment. Compared with existing PINN methods, the proposed FT-PINN method achieves a reduction in relative L_2 error by two to three orders of magnitude.
- The method demonstrates excellent effectiveness and applicability for aerothermoelastic analysis in a long duration under various loading conditions, different damping models, non-uniform geometric parameters, and time-varying parameters, accurately capturing a range of responses including thermal buckling, limit cycle oscillations (LCO), quasi-periodic and chaotic motions.

The structure of the paper is organized as follows. The aerothermoelastic problem of panels and the limitations of the vanilla PINN method for solving this problem are introduced in Sections 2. In Section 3, the newly proposed FT-PINN method incorporating a time-sequential training strategy and the random Fourier feature mapping technique is presented. In Section 4, numerical experiments are conducted on a two-dimensional panel to verify the validity, performance and advantages of the FT-PINN method for aerothermoelastic analysis. Finally, the contributions of this work are summarized in Section 5.

2. Problem statement

The present work focuses on the aerothermoelastic problem of panels on supersonic vehicles, a classic fluid–structure interaction problem. As shown in Fig. 1, the skin panel is subjected to substantial aerodynamic loads from supersonic airflow and thermal loads resulting from aerodynamic heating. The combined effects of inertial, elastic, and damping forces, together with aerodynamic and thermal loads, can lead to static or dynamic instability in the panel, resulting in various deformation and vibration responses. Due to the influence of airflow, chordwise bending deformation of the panel is of primary importance. To facilitate analysis, researchers often neglect spanwise bending and focus on the chordwise profile of the panel as the primary subject of investigation, as shown in Fig. 1(b), which is the well-known two-dimensional panel model commonly used for the aerothermoelastic problem [3,55–58].

The governing equation of this model is expressed as follows:

$$\begin{cases} \rho h \dfrac{\partial^2 w(x,t)}{\partial t^2} + c\dfrac{\partial w(x,t)}{\partial t} + D\dfrac{\partial^4 w(x,t)}{\partial x^4} + (N_{\mathrm{T}} - N_{\mathrm{G}})\dfrac{\partial^2 w(x,t)}{\partial x^2} \\ +\Delta p(x,t) = 0 \\ (x,t) \in (0,l) \times (0,T] \end{cases} \tag{1}$$

where $w(x,t)$ is the transverse displacement of the panel; T is the duration time; l is the chordwise length of the panel; ρ is the panel density; h is the panel thickness; c is the viscous damping coefficient; $D = Eh^3/12(1-\nu^2)$ is the bending stiffness; E is the elastic modulus; ν is the Poisson's ratio; and N_{T} is the in-plane force induced by thermal expansion. N_{G} is the in-plane force generated by geometric nonlinearity [59] and can be written as

$$N_{\mathrm{G}} = \frac{Eh}{2l(1-\nu^2)}\int_0^l \left(\frac{\partial w(x,t)}{\partial x}\right)^2 \mathrm{d}x \tag{2}$$

The aerodynamic load $\Delta p(x,t)$ can be presented by the piston theory as follows [1]:

$$\Delta p(x,t) = \frac{\rho_{\mathrm{a}} V^2}{\sqrt{Ma^2-1}}\left[\frac{\partial w}{\partial x} + \frac{Ma^2-2}{(Ma^2-1)V}\frac{\partial w}{\partial t}\right]\mathrm{d}x \tag{3}$$

where ρ_{a} and V are the density and speed of airflow, respectively; $Ma =$

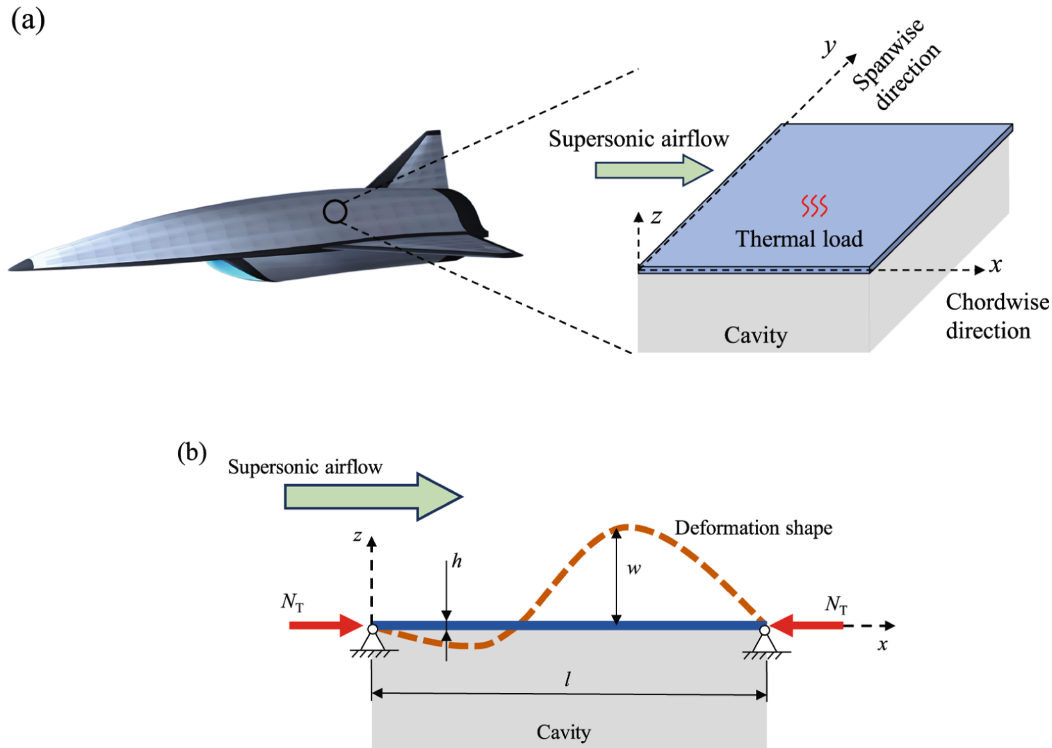

Fig. 1. Schematic illustration of a panel subjected to aerodynamic and thermal loads: (a) supersonic vehicle with three-dimensional panel model; and (b) two-dimensional panel model.

V/a_∞ is the Mach number; and a_∞ is the speed of sound.

Substituting Eqs. (2) and (3) into Eq. (1), and multiplying each term of Eq. (1) by l^4/Dh, Eq. (1) can be transformed to the following dimensionless form:

$$\begin{cases} \dfrac{\partial^2 W(\xi,\tau)}{\partial \tau^2} + (C+\zeta)\dfrac{\partial W(\xi,\tau)}{\partial \tau} + \dfrac{\partial^4 W(\xi,\tau)}{\partial \xi^4} \\ +\left[R_\mathrm{T} - 6\displaystyle\int_0^1 \left(\dfrac{\partial W(\xi,\tau)}{\partial \xi}\right)^2 \mathrm{d}\xi\right]\dfrac{\partial^2 W(\xi,\tau)}{\partial \xi^2} + \lambda \dfrac{\partial W(\xi,\tau)}{\partial \xi} = 0 \\ (\xi,\tau) \in (0,1)\times(0,\varGamma] \end{cases} \tag{4}$$

where $W = w/h$ is the dimensionless displacement; $\xi = x/l$ and $\tau = t\sqrt{D/\rho h l^4}$ are the dimensionless space and time coordinates, respectively; $\varGamma = T\sqrt{D/\rho h l^4}$ is the dimensionless duration time; $R_\mathrm{T} = N_\mathrm{T} l^2/D$ is the dimensionless thermally induced in-plane force; $C = c\sqrt{l^4/D\rho h}$ is the dimensionless viscous damping coefficient; λ and ζ are the dimensionless aerodynamic stiffness and damping, respectively, as follows:

$$\begin{cases} \lambda = \dfrac{\rho_\mathrm{a} V^2 l^3}{D\sqrt{Ma^2-1}} \\ \zeta = \dfrac{\rho_\mathrm{a} V (Ma^2-2)}{(Ma^2-1)^{3/2}}\sqrt{\dfrac{l^4}{D\rho h}} \end{cases} \tag{5}$$

Boundary and initial conditions are necessary for solving Eq. (4). For a simply supported panel, the boundary conditions are given by

$$\begin{cases} W(0,\tau) = W(1,\tau) = 0, & \tau \in [0,\varGamma] \\ \dfrac{\partial^2 W(0,\tau)}{\partial \xi^2} = \dfrac{\partial^2 W(1,\tau)}{\partial \xi^2} = 0, & \tau \in [0,\varGamma] \end{cases} \tag{6}$$

and the initial conditions are written as

$$\begin{cases} W(\xi,0) = h_1(\xi), & \xi \in (0,1) \\ \dfrac{\partial W(\xi,0)}{\partial \tau} = h_2(\xi), & \xi \in (0,1) \end{cases} \tag{7}$$

where $h_1(\xi)$ and $h_2(\xi)$ are the given initial displacement and velocity, respectively.

We can obtain the nonlinear aerothermoelastic response of the panel by solving Eq. (4), which is crucial for evaluate structural safety. Under different aerodynamic and thermal loads, the panel exhibits different types of aerothermoelastic response such as thermal buckling and flutter. Because of the effects of geometric nonlinearity, the panel displacement is not divergent but limited to a finite value. Therefore, the panel flutter is usually manifested as limit cycle oscillations (LCO), quasi-periodic and chaotic motions.

Accurately solving the aerothermoelastic responses from Eq. (4) is essential to study the aerothermoelastic problem. Recently, the rapid development of PINNs has introduced a new computational paradigm for solving governing equations in various engineering and scientific fields, e.g., fluid mechanics, solid mechanics, structural dynamics, and biomedical sciences. PINNs utilize neural networks to learn the mapping from spatiotemporal coordinates to solutions of PDEs. Physical laws, including PDEs, boundary conditions, and initial conditions, are incorporated into the loss function to guide the training process. When the loss function is minimized to a sufficiently low value, the network outputs can be considered approximate solutions to the PDEs. Developing an efficient PINN method for solving the nonlinear aerothermoelastic equation (4) is promising and appealing. However, the vanilla PINN approach proposed by Raissi et al. [22] has the following limitations and difficulties in addressing such a problem:

(i) **Integral term:** PINNs require the derivatives of the network outputs with respect to the network inputs. Based on the automatic differentiation technique [60], these derivatives can be easily and analytically calculated, so the vanilla PINN approach is suitable for solving PDEs. However, Eq. (4) is an integro-differential equation (IDE), and the nonlinear integral term $\int_0^1 \left(\frac{\partial W(\xi,\tau)}{\partial \xi}\right)^2 \mathrm{d}\xi$ cannot be presented analytically by the PINN codes. Although some numerical integration methods like Gaussian quadrature have been employed in PINN to compute the integral terms [61], it requires dense sample points in spatiotemporal domain. Moreover, numerical integration errors can accumulate and amplify when solving time-dependent aerothermoelastic problems, highlighting the need for more accurate methods to handle the integral term.

(ii) **Long-duration accurate solution:** Aerothermoelastic analysis typically requires accurately solving the governing equations over long-duration periods to obtain steady-state responses. As time progresses, the domain of the governing equations expands, making it increasingly challenging for the network to learn the mapping from the domain to the solution, and thus the vanilla PINN often fails to provide accurate solutions for long-duration problems. Furthermore, all terms of Eq. (4), including the aerodynamic load, are dependent on the deformation of the panel. This implies that the aerothermoelastic system lacks independent external excitation, resulting in a self-excited vibration where the instantaneous response is highly dependent on its previous states. Consequently, maintaining high solution accuracy throughout long-duration simulations is essential for reliable aerothermoelastic analysis.

(iii) **Nonlinear and multi-frequency properties:** Due to the effects of geometric nonlinearity, the aerothermoelastic response of the panel has strong nonlinear properties, leading to complex dynamic motions like multi-period LCO, quasi-periodic motion, and chaotic motion. These responses usually have multiple frequency components. However, according to the neural tangent kernel (NTK) theory [50,51], the fully-connected PINN tends to learn the low-frequency components, while has difficulty in learning high-frequency components. This phenomenon is commonly observed in PINN and called "***spectral bias***" [26,62,63]. It seriously prevents the vanilla PINN from accurately predicting the nonlinear and multi-frequency aerothermoelastic response.

3. Proposed method: FT-PINN

To address the above issues, we propose a novel Fourier time-sequential physics-informed neural network (FT-PINN) for the nonlinear aerothermoelastic analysis. The FT-PINN method converts Eq. (4) from an IDE to a set of PDEs by introducing a primitive function. A time-sequential training strategy, incorporating with a normalization technique and hard constraint, is employed to guarantee solution accuracy in a long duration. Furthermore, this method incorporates random Fourier feature mapping to preprocess input spatiotemporal coordinates, enhancing its ability to capture the nonlinear and multi-frequency characteristics of aerothermoelastic responses. The schematic diagram of the proposed FT-PINN method is illustrated in Fig. 2 and its details are presented as follows.

3.1. Transforming IDE to PDEs

Due to the in-plane force generated by geometric nonlinearity, the aerothermoelastic equation (4) is an IDE. Instead of calculating the nonlinear integral term by numerical integration methods, we introduce the following Φ function, which is defined as a primitive function of $\left(\frac{\partial W(\xi,\tau)}{\partial \xi}\right)^2$:

$$\Phi(\xi,\tau) = \int_0^\xi \left(\frac{\partial W(\xi,\tau)}{\partial \xi}\right)^2 \mathrm{d}\xi + \text{constant} \tag{8}$$

Therefore, the nonlinear integral term in Eq. (4) can be computed by the Φ function:

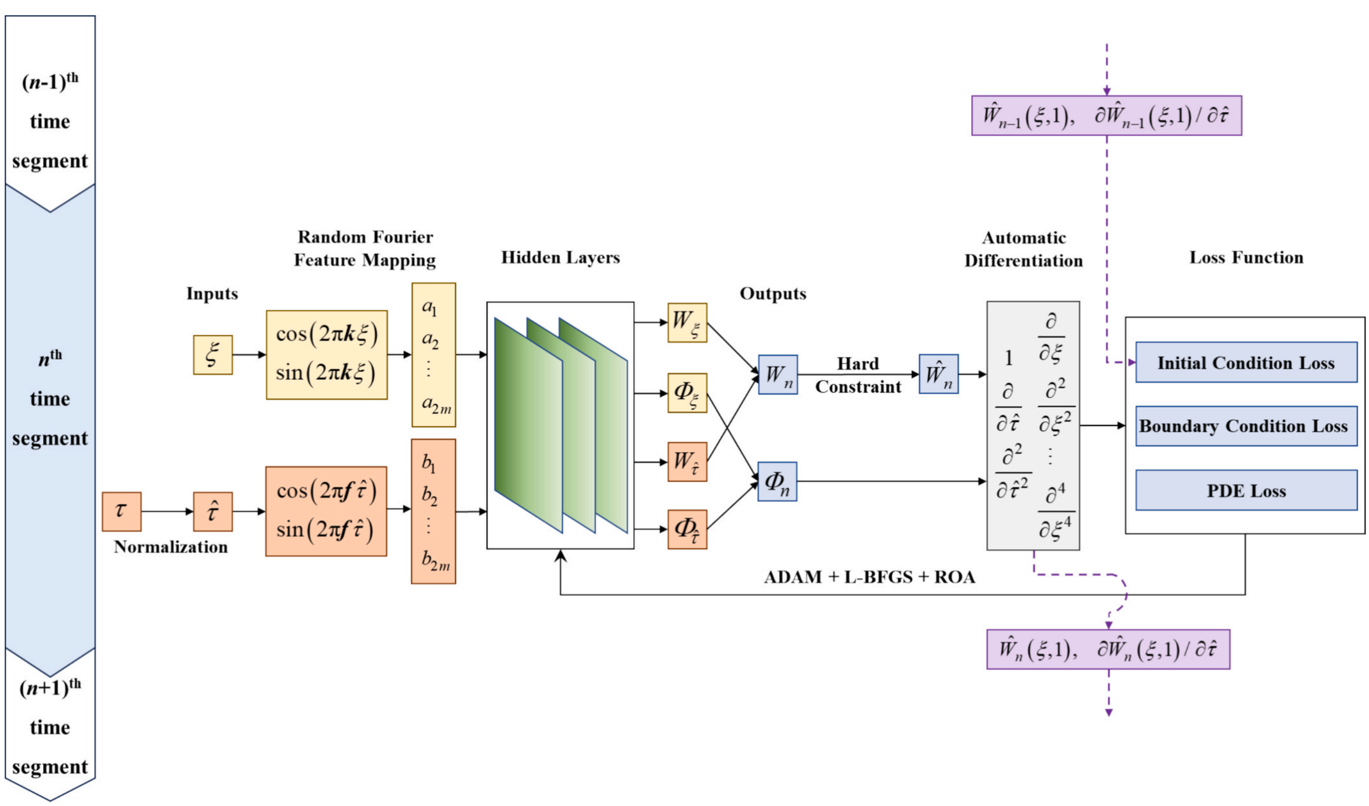

Fig. 2. Schematic diagram of the FT-PINN method.

$$\int_0^1 \left(\frac{\partial W(\xi,\tau)}{\partial \xi}\right)^2 \mathrm{d}\xi = \Phi(1,\tau) - \Phi(0,\tau) \tag{9}$$

In addition, a PDE describing the relationship between Φ and W can be derived from Eq. (8) as follows:

$$\frac{\partial \Phi(\xi,\tau)}{\partial \xi} = \left(\frac{\partial W(\xi,\tau)}{\partial \xi}\right)^2 \tag{10}$$

Substituting Eq. (9) into Eq. (4), we can convert Eq. (4) to the following system of PDEs:

$$\begin{cases} \dfrac{\partial^2 W(\xi,\tau)}{\partial \tau^2} + (C+\zeta)\dfrac{\partial W(\xi,\tau)}{\partial \tau} + \dfrac{\partial^4 W(\xi,\tau)}{\partial \xi^4} \\ +[R_\mathrm{T} - 6\Phi(1,\tau) + 6\Phi(0,\tau)]\dfrac{\partial^2 W(\xi,\tau)}{\partial \xi^2} + \lambda\dfrac{\partial W(\xi,\tau)}{\partial \xi} = 0 \\ \dfrac{\Phi(\xi,\tau)}{\partial \xi} = \left(\dfrac{\partial W(\xi,\tau)}{\partial \xi}\right)^2 \\ (\xi,\tau) \in (0,1)\times(0,\Gamma] \end{cases} \tag{11}$$

Equation (11) is equivalent to Eq. (4). By solving Eq. (11), the aerothermoelastic response can be obtained. Different from Eq. (4), all terms of Eq. (11) are differential, and they can be analytically calculated by the automatic differentiation technique. This treatment is inspired by the A-PINN method [64]. The use of A-PINN solves IDEs by introducing an auxiliary PDE and its boundary conditions. In contrast, our approach incorporates only an additional PDE, without the necessity of imposing additional boundary conditions.

3.2. Time-sequential training strategy

It is commonly observed in the vanilla PINN method that the solution error grows rapidly as time evolves [44–47]. This is mainly because the gradually increased time domain makes the mapping relationship from the time coordinates to the solutions more and more difficult to be learn by the neural network. To achieve long-duration and accurate aerothermoelastic analysis, we use a time-sequential training strategy in the present FT-PINN method. In this strategy, the whole time domain is decomposed to M time segments with the same interval $\Delta\Gamma$:

$$[0,\Gamma_1], [\Gamma_1,\Gamma_2], \cdots, [\Gamma_{n-1},\Gamma_n], \cdots, [\Gamma_{M-1},\Gamma] \tag{12}$$

where $[\Gamma_{n-1},\Gamma_n]$ is the n^{th} time segment and $\Gamma_n - \Gamma_{n-1} = \Delta\Gamma$. Since training the neural network over the entire time domain is difficult, we instead train it sequentially on individual time segments. After the network is trained in the n^{th} time segment, the solution of PDEs can be predicted by it, and then we can use the solution at the end of the n^{th} time segment as the initial conditions to train the $(n+1)^{\mathrm{th}}$ time segment. Because each time segment is much shorter than the whole time domain, it is easy to train the network to satisfy the given physics laws in each time segment. As a result, the loss function can be minimized to a sufficiently small value, which enables the attainment of satisfactory solutions. It is worth noting that, because the initial conditions of the $(n+1)^{\mathrm{th}}$ time segment are taken from the solutions in the n^{th} time segment, the errors of solutions will accumulate over these time segments. To ensure sufficient accuracy in long-duration aerothermoelastic analysis, it is essential to minimize solution errors within each time segment. To achieve this, the proposed FT-PINN method incorporates two key techniques, i.e., normalization and hard constraints.

Taking the n^{th} time segment as an example, where $\tau \in [\Gamma_{n-1},\Gamma_n]$. According to the min–max normalization method [65–67], we can get the normalized dimensionless time coordinate as

$$\widehat{\tau} = \frac{\tau - \tau_{\min}}{\tau_{\max} - \tau_{\min}} = \frac{\tau - \Gamma_{n-1}}{\Delta\Gamma} \tag{13}$$

As shown in Fig. 2, $\widehat{\tau}$ and ξ (the dimensionless space coordinate) are the inputs of FT-PINN, and they keep in the same range of $[0,1]$ for each time segment. Therefore, the same input data can be used to train the network across all time segments, this eliminates the need for resampling for each segment. Moreover, using $\widehat{\tau}$ instead of τ as the input can avoid the training difficulty induced by the increase of τ. Consequently, the loss function achieves a lower value when using $\widehat{\tau}$, which can improve solution accuracy.

Let W_n and Φ_n denote the outputs of FT-PINN. W_n and Φ_n are expect to be the approximate solutions of Eq. (11) in the n^{th} time segment, so they must satisfy the given physics laws including PDEs, boundary and initial conditions. According to the boundary conditions in Eq. (6), the boundary condition loss functions can be constructed as

$$\begin{cases} L_{b1}(\boldsymbol{\theta}) = \dfrac{1}{N_b}\sum_{k=1}^{N_b}\left[W_n\left(\xi_k^b,\widehat{\tau}_k^b\right)\right]^2, & \left(\xi_k^b,\widehat{\tau}_k^b\right) \in \{0,1\}\times[0,1] \\ L_{b2}(\boldsymbol{\theta}) = \dfrac{1}{N_b}\sum_{k=1}^{N_b}\left[\dfrac{\partial^2 W_n\left(\xi_k^b,\widehat{\tau}_k^b\right)}{\partial \xi^2}\right]^2, & \left(\xi_k^b,\widehat{\tau}_k^b\right) \in \{0,1\}\times[0,1] \end{cases} \tag{14}$$

where L_{b1} and L_{b2} are the boundary displacement and curvature losses, respectively; $\boldsymbol{\theta}$ is the trainable parameters of the neural network; $\left(\xi_k^b,\widehat{\tau}_k^b\right)$ and N_b are the coordinate and number of the sample points on boundaries, respectively. Embedding the boundary conditions into loss functions to constrain the training process of the neural network is called "soft constrains". Because the loss functions cannot converge to zero, the boundary conditions cannot be strictly satisfied, which is an important source of solution errors. To address this issue, we impose hard constraint to the network output W_n by introducing a trigonometric auxiliary function. The final output after hard constraining is expressed as

$$\widehat{W}_n(\xi,\widehat{\tau}) = \sin(\pi\xi)W_n(\xi,\widehat{\tau}) \tag{15}$$

Obviously, $\widehat{W}_n$ automatically meets the given boundary displacement in Eq. (6), so we only need to embed the other boundary conditions into the loss function. Therefore, the boundary loss function (i.e., the boundary curvature loss function) in terms of $\widehat{W}_n$ can be written as

$$L_b(\boldsymbol{\theta}) = \frac{1}{N_b}\sum_{k=1}^{N_b}\left[\frac{\partial^2 \widehat{W}_n\left(\xi_k^b,\widehat{\tau}_k^b\right)}{\partial \xi^2}\right]^2, \left(\xi_k^b,\widehat{\tau}_k^b\right) \in \{0,1\}\times[0,1] \tag{16}$$

For the system of PDEs in Eq. (11), considering $\frac{\partial^k}{\partial \tau^k} = \frac{\partial^k}{\partial \widehat{\tau}^k}\Delta\Gamma^{-k}$, the loss functions in terms of $\widehat{W}_n$ and Φ_n can be formulated as

$$\begin{cases} L_{p1}(\boldsymbol{\theta}) = \dfrac{1}{N_p}\sum_{k=1}^{N_p}\left\{\dfrac{\partial^2 \widehat{W}_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\Delta\Gamma^2\partial\widehat{\tau}^2} + (C+\zeta)\dfrac{\partial \widehat{W}_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\Delta\Gamma\partial\widehat{\tau}}\right. \\ +\dfrac{\partial^4 \widehat{W}_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\partial\xi^4} + \left[R_\mathrm{T} - 6\Phi_n\left(1,\widehat{\tau}_k^p\right) + 6\Phi_n\left(0,\widehat{\tau}_k^p\right)\right]\dfrac{\partial^2 \widehat{W}_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\partial\xi^2} \\ \left. +\lambda\dfrac{\partial \widehat{W}_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\partial\xi}\right\}^2 \\ L_{p2}(\boldsymbol{\theta}) = \dfrac{1}{N_p}\sum_{k=1}^{N_p}\left[\dfrac{\Phi_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\partial\xi} - \left(\dfrac{\partial \widehat{W}_n\left(\xi_k^p,\widehat{\tau}_k^p\right)}{\partial\xi}\right)^2\right]^2 \\ \left(\xi_k^p,\widehat{\tau}_k^p\right) \in (0,1)\times(0,1] \end{cases} \tag{17}$$

where L_{p1} and L_{p2} are the PDE losses for the first and second PDEs in Eq. (11), respectively; $\left(\xi_k^p,\widehat{\tau}_k^p\right)$ and N_p are the coordinate and number of the sample points over the normalized spatiotemporal domain, respectively.

According to the time-sequential training strategy, the initial conditions of the n^{th} time segment is taken from the solutions in the $(n-1)^{\mathrm{th}}$ time segment. As shown in Fig. 2, the initial loss functions for the n^{th} time segment are expressed as

$$\begin{cases} L_{i1}(\boldsymbol{\theta}) = \dfrac{1}{N_i}\sum_{k=1}^{N_i}\left[\widehat{W}_n\left(\xi_k^i, 0\right) - \widehat{W}_{n-1}\left(\xi_k^i, 1\right)\right]^2, & \xi_k^i \in (0,1) \\ L_{i2}(\boldsymbol{\theta}) = \dfrac{1}{N_i}\sum_{k=1}^{N_i}\left[\dfrac{\partial \widehat{W}_n\left(\xi_k^i, 0\right)}{\partial \widehat{\tau}} - \dfrac{\partial \widehat{W}_{n-1}\left(\xi_k^i, 1\right)}{\partial \widehat{\tau}}\right]^2, & \xi_k^i \in (0,1) \end{cases} \tag{18}$$

where L_{i1} and L_{i2} are the initial displacement and velocity losses, respectively; ξ_k^i and N_i are the space coordinate and number of the sample points at the $\widehat{\tau} = 0$, respectively; $\widehat{W}_{n-1}\left(\xi_k^i, 1\right)$ and $\partial \widehat{W}_{n-1}\left(\xi_k^i, 1\right)/\partial \widehat{\tau}$ are the displacement and velocity at the end of the (n-1)$^{\text{th}}$ time segment, i.e., the expected initial displacement and velocity of the n^{th} time segment. For the first time segment, $\widehat{W}_{n-1}\left(\xi_k^i, 1\right)$ and $\partial \widehat{W}_{n-1}\left(\xi_k^i, 1\right)/\partial \widehat{\tau}$ are replaced with the given $h_1\left(\xi_k^i\right)$ and $h_2\left(\xi_k^i\right)\Delta\Gamma$ as shown Eq. (7). It is possible to impose hard constraints to initial conditions, but it may lead to more complex mapping relationship for the neural network to learn, and thus may cause adverse effects to the final solution accuracy [48]. Therefore, we still employ soft constrains to embed the initial conditions in the loss function.

Finally, the total loss function L of the FT-PINN method is made up of the above boundary, PDE and initial losses:

$$L(\boldsymbol{\theta}) = L_{p1}(\boldsymbol{\theta}) + \lambda_{p2}L_{p2}(\boldsymbol{\theta}) + \lambda_b L_b(\boldsymbol{\theta}) + \lambda_{i1}L_{i1}(\boldsymbol{\theta}) + \lambda_{i2}L_{i2}(\boldsymbol{\theta}) \tag{19}$$

where λ_{p2}, λ_b, λ_{i1} and λ_{i2} are the weights of these loss components, which balance the effects of different physics laws.

3.3. Random Fourier feature mapping

The nonlinear aerothermoelastic behavior of panels often involves multiple frequency components, particularly during the analysis of panel flutter. However, the fully-connected neural networks typically used in PINN methods struggle to capture high-frequency components due to spectral bias, leading to suboptimal performance in multi-frequency scenarios. To overcome this limitation, we integrate random Fourier feature mapping into the FT-PINN method, which transforms spatiotemporal coordinates into a set of sine and cosine functions. As shown in Fig. 2, because the vibration frequency (in time domain) and the wavenumber (in space domain) are generally different, inspired by the separation of variables, we perform the random Fourier feature mapping technique on the space and time coordinates, respectively:

$$\begin{cases} a(\xi) = \begin{bmatrix} \cos(2\pi \boldsymbol{k}\xi) \\ \sin(2\pi \boldsymbol{k}\xi) \end{bmatrix} \\ b(\widehat{\tau}) = \begin{bmatrix} \cos(2\pi \boldsymbol{f}\widehat{\tau}) \\ \sin(2\pi \boldsymbol{f}\widehat{\tau}) \end{bmatrix} \end{cases} \tag{20}$$

where $\boldsymbol{a} = [a_1, a_2, \cdots, a_{2m}]^{\mathrm{T}}$ and $\boldsymbol{b} = [b_1, b_2, \cdots, b_{2m}]^{\mathrm{T}}$ are the Fourier feature mapping values; $\boldsymbol{k} = [k_1, k_2, \cdots, k_m]^{\mathrm{T}}$ and $\boldsymbol{f} = [f_1, f_2, \cdots, f_m]^{\mathrm{T}}$ are the mapping factors; m is the number of mapping dimension. Here, $\boldsymbol{k}$ and $\boldsymbol{f}$ are in fact the nondimensional wavenumber and frequency vectors, respectively. Their elements are randomly sampled from Gaussian distributions $\mathcal{N}\left(0, \sigma_\xi^2\right)$ and $\mathcal{N}\left(0, \sigma_{\widehat{\tau}}^2\right)$, respectively, where σ_ξ and $\sigma_{\widehat{\tau}}$ are standard deviations.

Let $\Psi(\boldsymbol{x}, \boldsymbol{\theta})$ denote a fully-connected neural network with trainable parameters $\boldsymbol{\theta}$, $2m$ inputs, and 2 outputs. Instead of inputting ξ and $\widehat{\tau}$ to this network, we input the Fourier feature mapping values $\boldsymbol{a}$ and $\boldsymbol{b}$ to the network, respectively, and get the outputs

$$\begin{cases} \begin{bmatrix} W_\xi \\ \Phi_\xi \end{bmatrix} = \Psi(\boldsymbol{a}(\xi), \boldsymbol{\theta}) \\ \begin{bmatrix} W_{\widehat{\tau}} \\ \Phi_{\widehat{\tau}} \end{bmatrix} = \Psi(\boldsymbol{b}(\widehat{\tau}), \boldsymbol{\theta}) \end{cases} \tag{21}$$

where W_ξ and Φ_ξ are the network outputs corresponding to the space coordinates ξ, while $W_{\widehat{\tau}}$ and $\Phi_{\widehat{\tau}}$ are the network outputs corresponding to the time coordinates $\widehat{\tau}$. Note that $\boldsymbol{a}$ and $\boldsymbol{b}$ are passed through the same network, and $\boldsymbol{k}$ and $\boldsymbol{f}$ are kept fixed during training process, so the FT-PINN method does not require more trainable parameters than the PINN methods with a conventional fully-connected neural network. Then, the outputs W_n and Φ_n of FT-PINN in the n^{th} time segment before hard constraining are computed by multiplying the corresponding components together:

$$\begin{cases} W_n = W_\xi W_{\widehat{\tau}} \\ \Phi_n = \Phi_\xi \Phi_{\widehat{\tau}} \end{cases} \tag{22}$$

Finally, after applying the hard constraint strategy according to Eq. (15), the final output $\widehat{W}_n$ of FT-PINN can be obtained. Reviewing this section, we found that the random Fourier feature mapping technique constructs multiple waves with different wavenumbers and frequencies. The outputs of FT-PINN are the nonlinear functions of these waves, so it can significantly enhance the ability of FT-PINN to capture the multi-frequency properties of aerothermoelastic behavior. It is important to note that the multi-frequency properties are intrinsically linked to the nonlinearity of the aerothermoelastic system. Going beyond the fundamental-frequency response, nonlinearity can generate higher-order harmonics, consequently leading to multi-frequency responses. Although the vanilla PINN approach with nonlinear activation functions can model nonlinear systems, it suffers from spectral bias, favoring the learning of low-frequency responses and struggling to capture higher-order responses. In contrast, the random Fourier feature mapping technique in FT-PINN ensures effective representation of multiple frequencies, thereby enhancing its capacity to capture the nonlinear behavior of the aerothermoelastic system. In addition, adjusting the mapping dimension m and the standard deviations σ_ξ and $\sigma_{\widehat{\tau}}$ allows FT-PINN to tune its frequency learning to a desired range.

To address the problem of error accumulation over time for aerothermoelastic analysis, we embed the random Fourier feature mapping technique [26,53,68] into a time-sequential training architecture, and this allows FT-PINN to provide accurate solutions for nonlinear and multi-frequency problems over long-duration periods. The detailed algorithm of the FT-PINN approach is presented in Appendix A. In the next section, we will quantitatively compare FT-PINN with existing methods to highlight its advantages through a series of numerical experiments.

4. Results and discussion

In this section, we demonstrate the effectiveness of the proposed FT-PINN method by applying it to the nonlinear aerothermoelastic problem described in Section 2. Furthermore, we conduct aerothermoelastic analyses of panels under various loading conditions, with different damping models, non-uniform thickness and time-varying parameters to verify the applicability and robustness of FT-PINN.

As described in Section 3, in the following numerical experiments, the neural network architecture of FT-PINN has 2 inputs (ξ and $\widehat{\tau}$) and 2 outputs ($\widehat{W}_n$). The dimension numbers of the random Fourier feature mapping technique for space and time coordinates are both set to be $m = 128$. A fully-connected neural network with 256 input neurons and 2 output neurons follows the random Fourier feature mapping. And it has 2 hidden layers, each containing 256 neurons. The sine activation function is used in the neural network due to its strong performance in dynamic problems [47]. The network is initially pre-trained using the Adam optimizer for 1,500 iterations, followed by training with a modified L-BFGS optimizer that incorporates a reactivating optimization algorithm (ROA) [48]. This algorithm helps avoid local optima and enhances solution accuracy. All numerical simulations in this section are performed on an NVIDIA RTX 4090 GPU with 24 GB of memory.

Table 1
Relative L_2 errors and computational costs of different methods for aerothermoelastic analysis.

Method	Vanilla PINN [22]	MscaleDNN [52,69]	ST-MFF-PINN [26,53]	AT-PINN-HC [48]	FT-PINN (This work)
Relative L_2 error	1.12E+00	1.06E+00	1.00E+00	2.70E-01	5.55E-03
Training time per 100 iterations (s)	1.91	5.48	4.17	1.98	4.22
Maximum allocated memory (GB)	1.21	3.65	2.29	1.21	2.29

4.1. Verification of the proposed FT-PINN method

The aerothermoelastic response of a simply-supported panel as described in Section 2 is calculated by the proposed FT-PINN method in this section. The panel is made of titanium alloy, and the material parameters are $E = 110$ GPa, $\nu = 0.34$ and $\rho = 4400$ kg/m^3. The geometric parameters of the panel are $h = 2$ mm and $l = 1$ m. The viscous damping coefficient is set to $c = 10$ N·s/m^3. The parameters related to the thermal and aerodynamic loads are as follows: $N_\mathrm{T} = 2500$ N/m, $\rho_a = 0.041$ kg/m^3, $a_\infty = 295$ m/s and $Ma = 3$. Based on these dimensional parameters, the dimensionless parameters of Eq. (4) can be computed as: $C = 0.37$, $R_\mathrm{T} = 30.2$, $\lambda = 136.9$, $\zeta = 0.416$. The initial displacement is given by $h_1(\xi) = \sin(\pi\xi)$, and the initial velocity is set to zero. The dimensionless simulation duration is selected as $\Gamma = 30$ to verify the performance of FT-PINN for a long duration and guarantee steady-state aerothermoelastic responses can be observed.

Using FT-PINN, the entire time domain is split with a time interval of $\Delta\Gamma = 0.15$. The numbers of sample points in each time segment for calculating PDE losses, boundary and initial condition losses are selected as $N_p = 5000$, $N_b = 200$ and $N_i = 50$, respectively. The weights of PDE, boundary and initial condition losses are set to $\lambda_{p2} = 1$, $\lambda_b = 10$, $\lambda_{i1} = 10000$ and $\lambda_{i2} = 10$. The standard deviations of the random Fourier feature mapping technique for the space and time coordinates are set to $\sigma_\xi = 1$ and $\sigma_{\hat{\tau}} = 2$, respectively. The hyperparameters of FT-PINN are determined through a trial-and-error process and some representative results are presented in Appendix B. To quantitatively assess the accuracy of FT-PINN, the aerothermoelastic response is computed using the traditional Galerkin method with Runge–Kutta integration, which serves as the reference solution. The relative L_2 error of FT-PINN is calculated as follows:

$$\varepsilon = \frac{\sqrt{\frac{1}{N_r}\sum_{k}^{N_r}\left[\widehat{W}\left(\xi_k^r, \tau_k^r\right) - W_{ref}\left(\xi_k^r, \tau_k^r\right)\right]^2}}{\sqrt{\frac{1}{N_r}\sum_{k}^{N_r}\left[W_{ref}\left(\xi_k^r, \tau_k^r\right)\right]^2}}, \left(\xi_k^r, \tau_k^r\right) \in (0, 1) \times (0, \Gamma) \tag{23}$$

where $\widehat{W}$ and W_{ref} denote the FT-PINN and reference solutions, respectively; $\left(\xi_k^r, \tau_k^r\right)$ is the coordinate of points uniformly sampled in the whole spatiotemporal domain, and its number is $N_r = 201 \times 10{,}001 = 2{,}010{,}201$.

To better demonstrate the performance of the proposed FT-PINN method, some relevant PINN methods are applied to solve the aerothermoelastic problem, including the vanilla PINN [22], MscaleDNN [52,69], ST-MFF-PINN [26,53] and AT-PINN-HC [48] methods. Among these methods, the AT-PINN-HC method was developed by the authors for long-duration structural dynamic analysis, and both MscaleDNN and ST-MFF-PINN were developed to deal with multiscale and multi-frequency problems. MscaleDNN uses multiple neural networks with scaled inputs, while ST-MFF-PINN employs the random Fourier feature mapping technique. In this example, FT-PINN and ST-MFF-PINN have the same fully-connected neural network and random Fourier feature mapping dimension. The vanilla PINN and AT-PINN-HC methods do not use the random Fourier feature mapping technique but incorporate a larger fully-connected neural network with one additional hidden layer of 256 neurons. MscaleDNN employs three fully-connected neural networks, each identical to the one used in the vanilla PINN. The relative L_2 errors and computational costs of these methods are listed in Table 1. The aerothermoelastic responses of the panel predicted by these methods are illustrated in Fig. 3 and compared with the reference solution. The responses in Fig. 3 are selected from the typical point at $\xi = 0.7$, because the maximum aerothermoelastic response usually appears around this point.

From Table 1 and Fig. 3(a), it can be seen that the vanilla PINN, MscaleDNN and ST-MFF-PINN methods cannot predict a satisfactory result for the long-duration aerothermoelastic analysis. As time evolves, the solution of the vanilla PINN method tends to zero. This phenomenon has also been observed in long-duration free vibration [47,48]. Although the solutions of MscaleDNN and ST-MFF-PINN show more fluctuations, their overall trends also converge to zero. This is because zero solution satisfies the given PDEs and boundary conditions for such a dynamic system without any external load independent to the system motion. Although zero solution violates the given nonzero initial conditions at $\tau = 0$, these initial conditions primarily constrain the neural network near $\tau = 0$ because the network is trained over the entire spatiotemporal domain simultaneously. As a result, the solutions of the vanilla PINN, MscaleDNN and ST-MFF-PINN methods are nonzero near $\tau = 0$ and gradually converge to zero, which cannot conform to the true solution. Noted that ST-MFF-PINN can also provide satisfactory results for long-duration simulations by utilizing plenty of known data inside the spatiotemporal domain [53]. However, these data are generally unavailable especially for forward problems. And the results shown here demonstrate that ST-MFF-PINN is not suitable for long-duration aerothermoelastic analysis without known data.

In comparison, the AT-PINN-HC method predicts a better solution, but noticeable discrepancies remain when compared to the reference solution as shown in Fig. 3(b). As demonstrated in Fig. 3(c), the solution obtained with the present FT-PINN method closely aligns with the reference solution across the entire time domain. This is further evidenced by a significantly low relative L_2 error of 5.55E-03, which is lower than other methods by 2 to 3 orders of magnitude. Both AT-PINN-HC and FT-PINN employ a time-sequential training strategy, this allows the initial conditions of each time segment to guide neural network training and prevent the solution from converging to an unreasonable zero. It is worth noting that the previous ST-MFF-PINN method usually requires multiple random Fourier feature mappings for space or time coordinates. In contrast, we only apply one random Fourier feature mapping to the space and time coordinates in FT-PINN and the results are satisfactory. The achievement is also attributed to the time-sequential training strategy, which effectively reduces the learning complexity for nonlinear and multi-frequency responses in each time segment.

Table 1 summarizes the computational costs of the vanilla PINN, MscaleDNN, ST-MFF-PINN, AT-PINN-HC and FT-PINN for this example, in terms of training time for performing 100 iterations and the maximum allocated memory. Among all methods, the vanilla PINN requires the least computational cost, closely followed by AT-PINN. Both ST-MFF-PINN and FT-PINN incur computational cost that is roughly double that of the vanilla PINN. This higher cost results from the separate input of space and time coordinates to the neural network, necessitating additional resources for computing the outputs and their derivatives. In comparison, MscaleDNN has the highest cost as it employs three independent fully-connected neural networks. In general, FT-PINN significantly enhances the solution accuracy for aerothermoelastic analysis at a

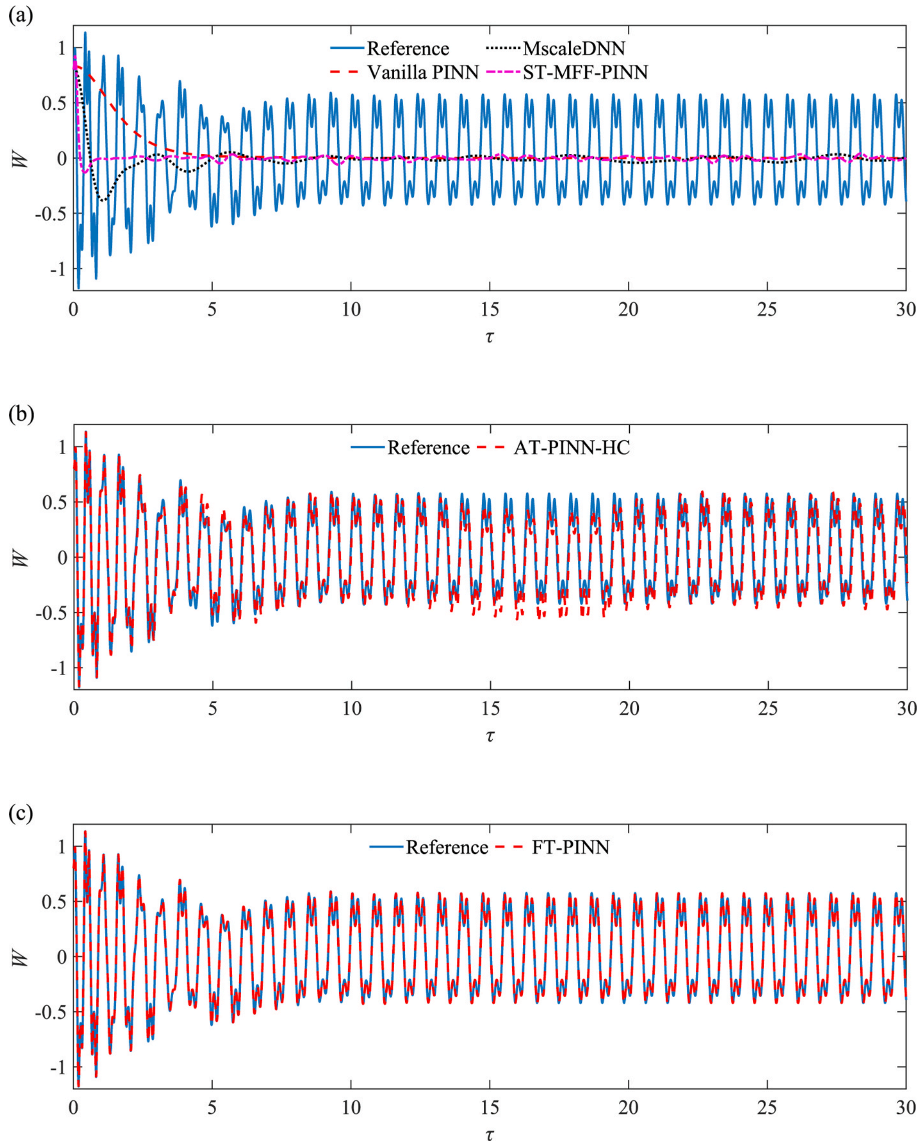

Fig. 3. Comparison of the performance of vanilla PINN, MscaleDNN, ST-MFF-PINN, AT-PINN-HC, and FT-PINN for aerothermoelastic analysis. The curves show the time histories of displacement at $\xi = 0.7$: (a) reference solution versus solutions from vanilla PINN, MscaleDNN, and ST-MFF-PINN; (b) reference solution versus AT-PINN-HC; and (c) reference solution versus FT-PINN.

reasonable training efficiency compared to the existing PINN methods. In addition, we compare the computational cost of FT-PINN against the traditional Galerkin method coupled with Runge–Kutta integration. The latter method is employed in this study for generating reference solutions. As indicated in Table 2, while FT-PINN requires minor time and memory to predict the aerothermoelastic response in the entire spatio-temporal domain using the trained network, its training process incurs substantial computation time, exceeding that of the Galerkin method. The prohibitively long training time remains a common drawback of PINN-based methods compared to traditional numerical methods. Significant enhancements in training efficiency are essential for PINNs to become a viable replacement in practical applications in the future.

Table 2
Computational costs between FT-PINN and Galerkin method with Runge–Kutta integration for aerothermoelastic analysis.

Method	Galerkin method with Runge–Kutta integration	FT-PINN (training)	FT-PINN (prediction)
Computation time	0.76 h	19.93 h	0.89 s
Maximum allocated memory (GB)	0.44	2.29	0.31

Furthermore, we compare the representative convergence plots of the loss functions for the vanilla PINN, MscaleDNN, ST-MFF-PINN, AT-PINN-HC and FT-PINN methods in Fig. 4. The five loss components of the proposed FT-PINN method are also shown in Fig. 4. As illustrated in Fig. 4(a), the loss function of FT-PINN converges faster than other methods and reaches the lowest value. It indicates that the random Fourier feature mapping technique improves the ability of FT-PINN to learn the nonlinear and multi-frequency properties of the aerothermoelastic system.

To further reveal the advantages of FT-PINN over AT-PINN-HC and the effects of the random Fourier feature mapping technique, some detailed results including the frequency spectrum, phase plot and Poincaré map of the steady-state aerothermoelastic response are presented in Fig. 5 and Fig. 6. From the frequency spectrum, the instantaneous deformation shape, the phase plot, the Poincaré map and the displacement contour plots, the excellent accuracy of FT-PINN is further verified, while AT-PINN-HC shows significant errors. According to the phase plot and the Poincaré map, the aerothermoelastic response of this example is a triple-period LCO, which is a typical nonlinear dynamic

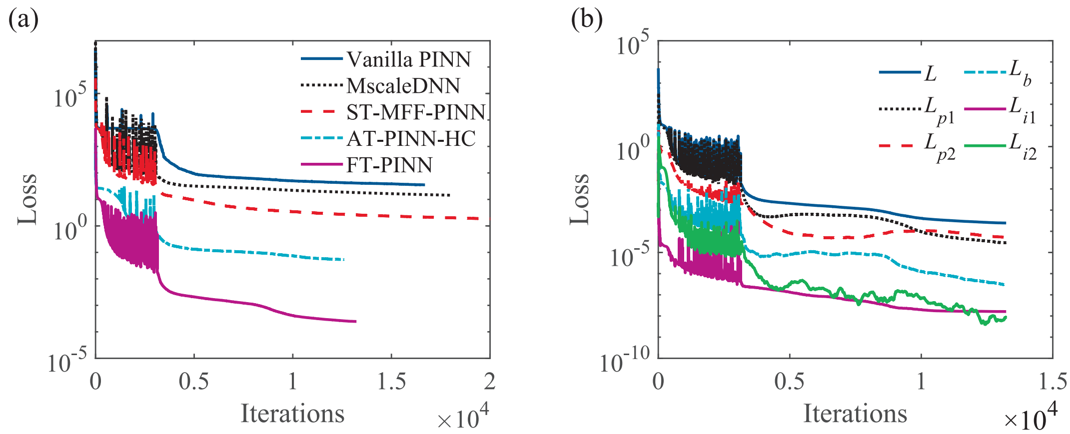

Fig. 4. Convergence plots of loss functions: (a) the total loss functions of vanilla PINN, MscaleDNN, ST-MFF-PINN, AT-PINN-HC, and FT-PINN; and (b) the total loss function of FT-PINN and its components in the first time segment.

phenomenon. From the frequency spectrum shown in Fig. 5(a), it can be seen that the nonlinear response contains a strong fundamental frequency of 3.9 along with its 2nd to 7th harmonics showing a multi-frequency property. The random Fourier feature mapping technique makes FT-PINN well capture the nonlinear and multi-frequency properties of the aerothermoelastic response. In contrast, AT-PINN-HC without the random Fourier feature mapping technique can only learn the response below the frequency of 5.0 and at some isolated frequency points of the 3rd and 5th harmonics. In addition, we analyze the evolution of the relative L_2 error with time to compare the error cumulation in AT-PINN-HC and FT-PINN. As illustrated in Fig. 5(e), both the relative L_2 errors of AT-PINN-HC and FT-PINN generally increase with time. While the relative L_2 error of AT-PINN-HC exceeds 1.0E-01 beyond $\tau = 6$, that of FT-PINN remains below 1.0E-02 throughout the entire duration. This demonstrates the capability of FT-PINN to provide accurate long-term solutions, which is crucial for capturing the steady-state response of aerothermoelastic problems.

4.2. Aerothermoelastic analysis under various loading conditions

In this section, we apply the FT-PINN method to analyze the aerothermoelastic response of panels under various thermal and aerodynamic loads. The dimensional and dimensionless loading parameters of 6 cases are listed in Table 3. The types of the aerothermoelastic responses in the 6 cases and the relative L_2 errors of FT-PINN are also summarized in Table 3. Some representative results are compared with the reference solutions in Figs. 7–11. Note that the Case 4 in Table 3 has been studied in Section 4.1 and thus the results are not redisplayed in this section. Supplementary information is also presented in Appendix C.

In Case 1, as shown in Fig. 7, the panel is stable under small thermal and aerodynamic loads. After the initial perturbation, its response gradually converges to zero, just like the damped free vibration. In Case 2, as shown in Fig. 8, the panel undergoes static instability (i.e., thermal buckling) under a small aerodynamic load but a heavy thermal load. After decaying oscillation, the response ultimately converges to a nonzero position. The final buckling configuration is also presented in Fig. 8(b).

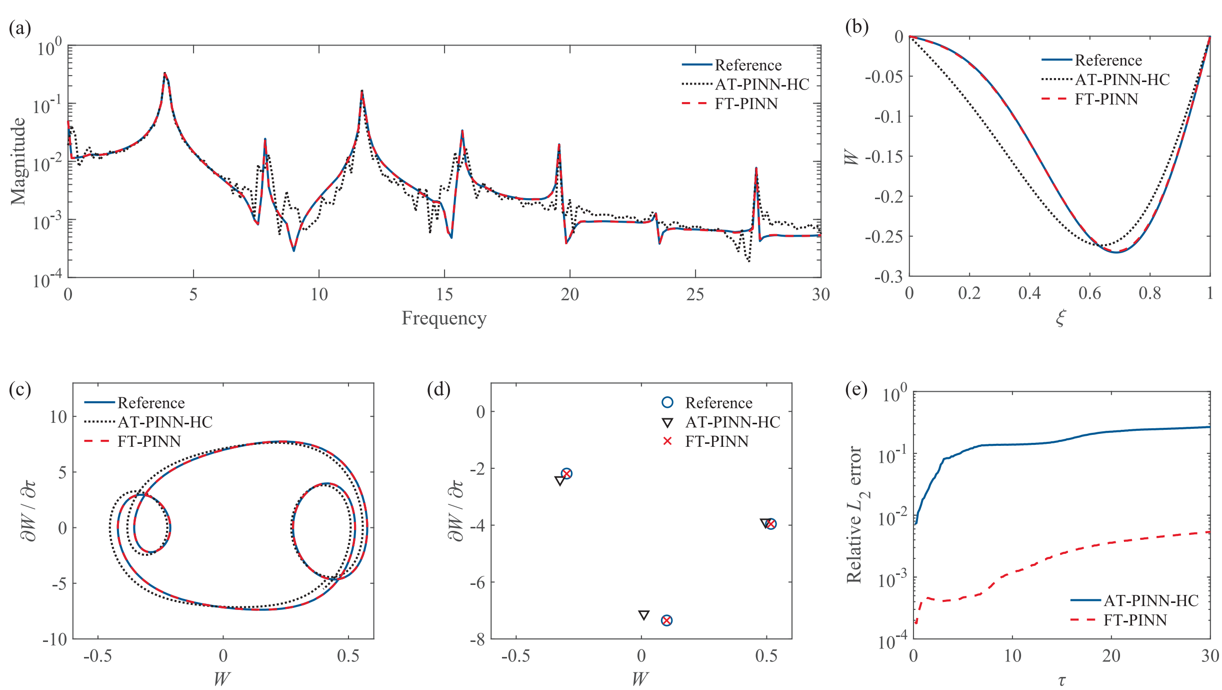

Fig. 5. Comparison of the performance of AT-PINN-HC and FT-PINN methods for aerothermoelastic analysis: (a) frequency spectrum; (b) instantaneous deformation shape at $\tau = 20$; (c) phase plot; (d) Poincaré map; and (e) evolution of the relative L_2 error.

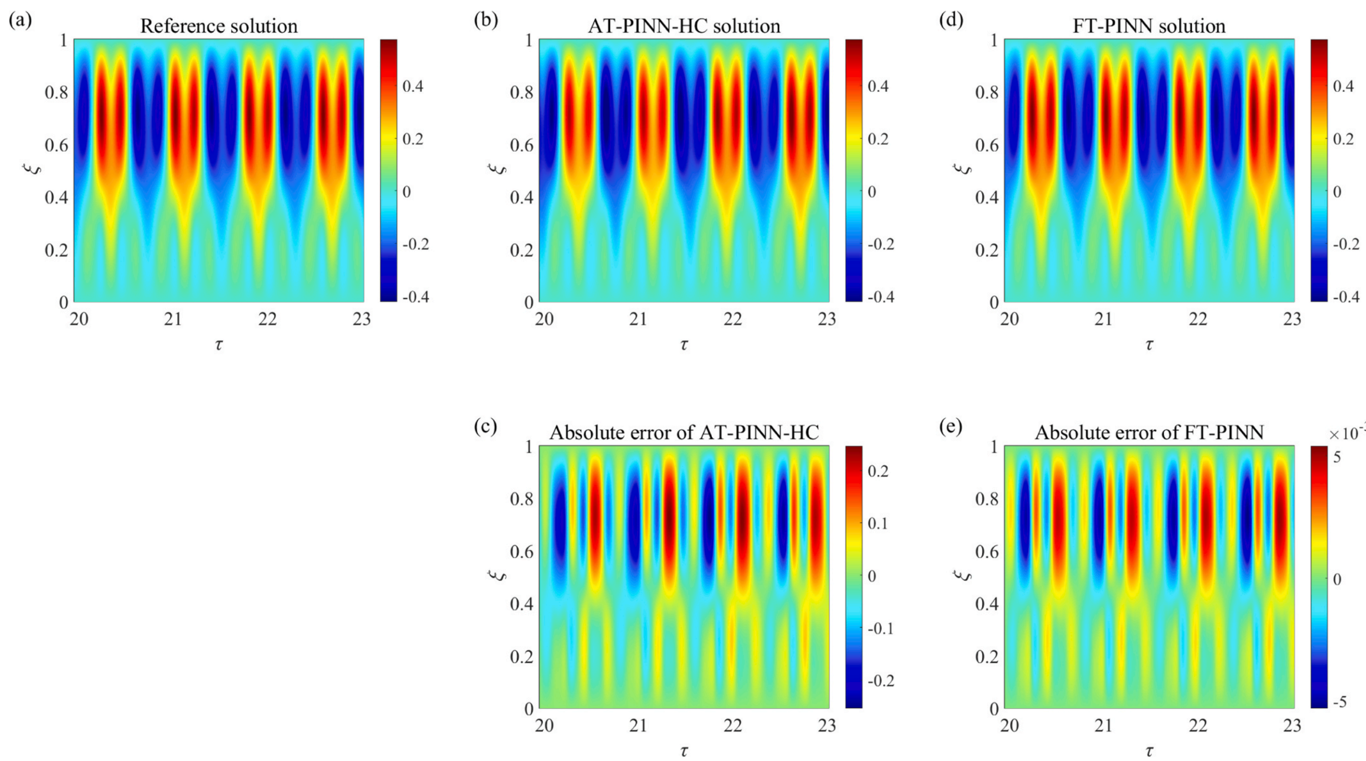

Fig. 6. Aerothermoelastic analysis results for $\tau \in [20, 23]$: (a) reference solution; (b)–(c) predicted solution and absolute error of AT-PINN-HC; and (d)–(e) predicted solution and absolute error of FT-PINN.

Table 3
Loading parameters, aerothermoelastic response types and relative L_2 errors of FT-PINN under various loading conditions.

Case No.	N_T (N/m)	ρ_a (kg/m^3)	a_∞ (m/s)	Ma	R_T	λ	ζ	Aerothermoelastic response type	Relative L_2 error
1	0	0.041	295	2	0	99.4	0.345	Convergence to zero	3.08E-04
2	2000	0.041	295	1.5	24.1	86.6	0.120	Thermal buckling	3.17E-03
3	0	0.089	295	4	0	385.9	0.937	Flutter (Single-period LCO)	3.82E-03
4	2500	0.041	295	3	30.2	136.9	0.416	Flutter (Triple-period LCO)	5.55E-03
5	3000	0.089	295	2	36.2	215.7	0.748	Flutter (Quasi-periodic motion)	1.40E-03
6	4000	0.089	295	2	48.2	215.7	0.748	Flutter (Chaotic motion)	1.31E+00

In Cases 3–6 as shown in Figs. 3, 5, 9, 10 and 11, the panel becomes dynamically unstable and vibrates with a limited magnitude, which is called panel flutter. Under different combinations of thermal and aerodynamic loads, the panel flutter exhibits different characteristics. In Case 3, as shown in Fig. 9, the vibration response is periodic, as evidenced by a simple closed cycle in the phase plot and a single point in the Poincaré map, indicating a single-period LCO. In Fig. 9(e), the response of Case 3 mainly contains a strong fundamental frequency of 17.0 along with its 3rd harmonic. As has been described in Figs. 3 and 5 in Section 4.1, the response of Case 4 is also periodic, but its Poincaré map has three points, so it is a triple-period LCO. Obviously, the response in Case 4 contains more frequencies than Case 3. In Case 5, it is hard to say whether the response is periodic or not from Fig. 10(a) and (c), but the Poincaré map in Fig. 10(d) is a closed curve, indicating that the response is a quasi-periodic motion. As shown in Fig. 10(e), the quasi-periodic response has richer frequency content than in Cases 3 and 4. In Case 6, as illustrated in Fig. 11, the response is random and non-periodic. Both phase plot and Poincaré map indicate that the motion is chaotic.

In Cases 1–5, the FT-PINN method consistently produces satisfactory results with relative L_2 errors less than 6.0E-03. This robustness is evident regardless of whether the aerothermoelastic response decays to zero, converges to a non-zero equilibrium, or exhibits LCO or quasi-periodic motion. Even in Cases 4 and 5, which exhibit complex frequency content, the solutions obtained by FT-PINN closely match the reference results. However, in Case 6, significant discrepancies are observed between the FT-PINN and reference solutions. This is mainly caused by the nature of chaotic motion. In a chaotic system, any initial minor errors can be exponentially amplified and thus lead to substantial differences as time progresses. In addition, exact or analytical solutions for a chaotic motion are generally not accessible. The reference solution presented herein is computed by the traditional Galerkin method combined with Runge–Kutta integration. Due to the inevitable error of this numerical method and the amplification effect of errors in the chaotic system, the reference solution also deviates considerably from the exact solution of chaotic motion. Therefore, the point-to-point errors shown in Fig. 11 **(a)-(d)** and the relative L_2 error of Case 6 listed in Table 3 are indeed not very meaningful to evaluate the performance of FT-PINN. Nevertheless, the FT-PINN solution qualitatively demonstrates that the system in Case 6 enters a chaotic state which is in accordance with the reference solution. Furthermore, we can quantitatively compare the FT-PINN and reference solutions through statistical metrics, such as power spectrum density, probability density and mean square value. Fig. 11 **(e)-(h)** shows the highly similar power spectrum densities and probability densities of the FT-PINN and reference solutions. The mean square values of the FT-PINN and reference solutions are 0.515 and 0.514, respectively, indicating a minor relative error of 1.95E-03. These comparative results demonstrate the effectiveness of FT-PINN for the aerothermoelastic analysis of a chaotic system.

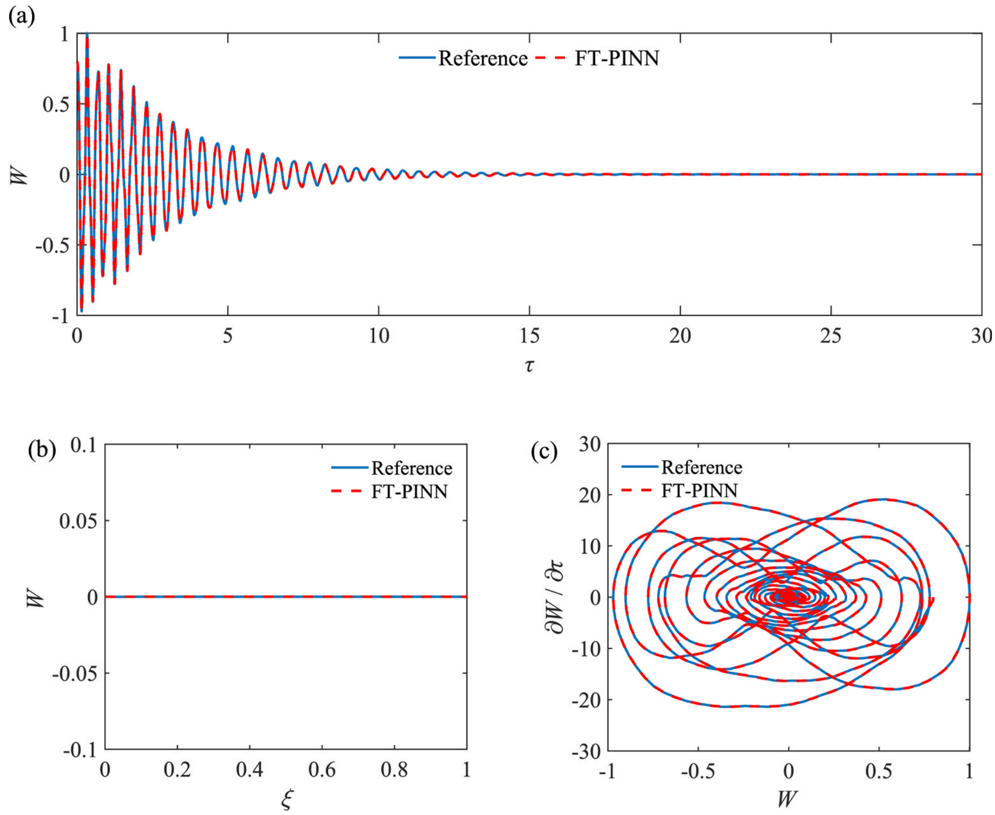

Fig. 7. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis under loading Case 1: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 30$; and (c) phase plot.

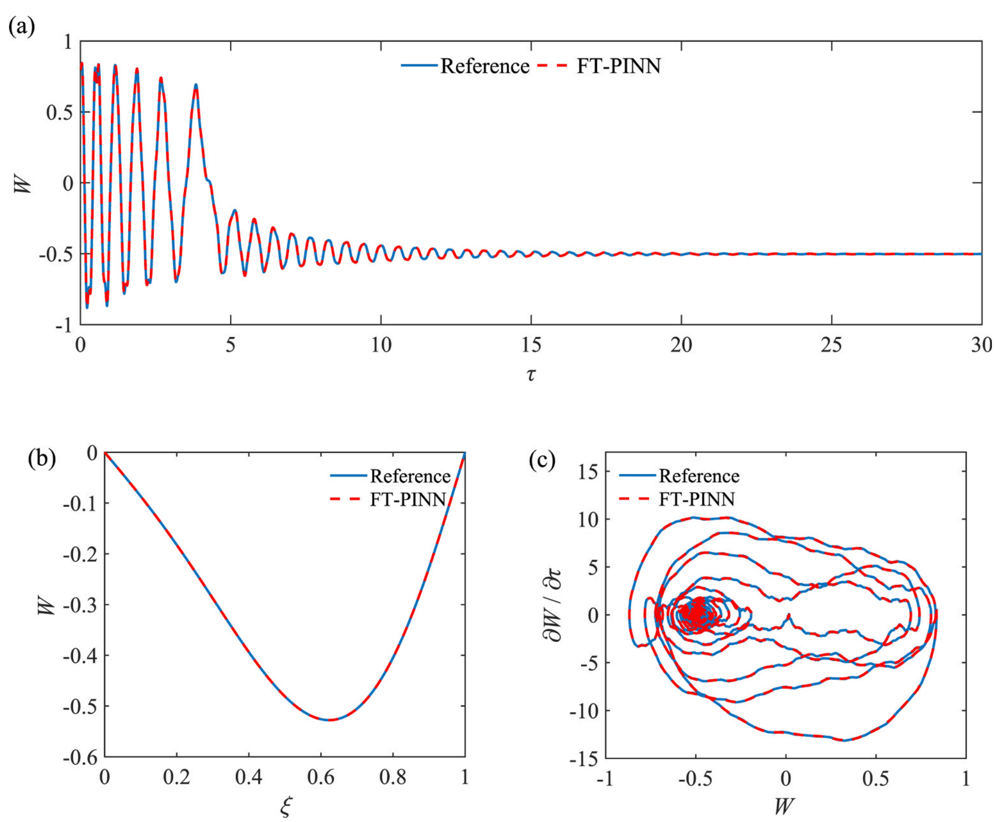

Fig. 8. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis under loading Case 2: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 30$; and (c) phase plot.

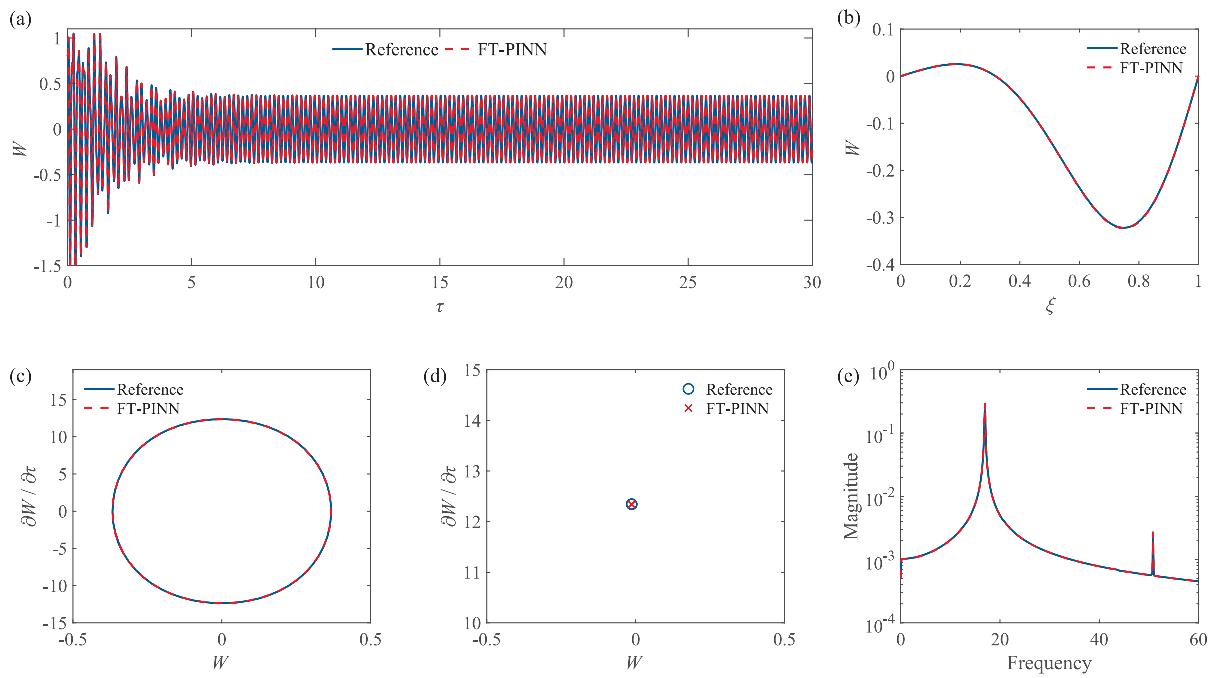

Fig. 9. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis under loading Case 3: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 20$; (c) phase plot; (d) Poincaré map; and (e) frequency spectrum.

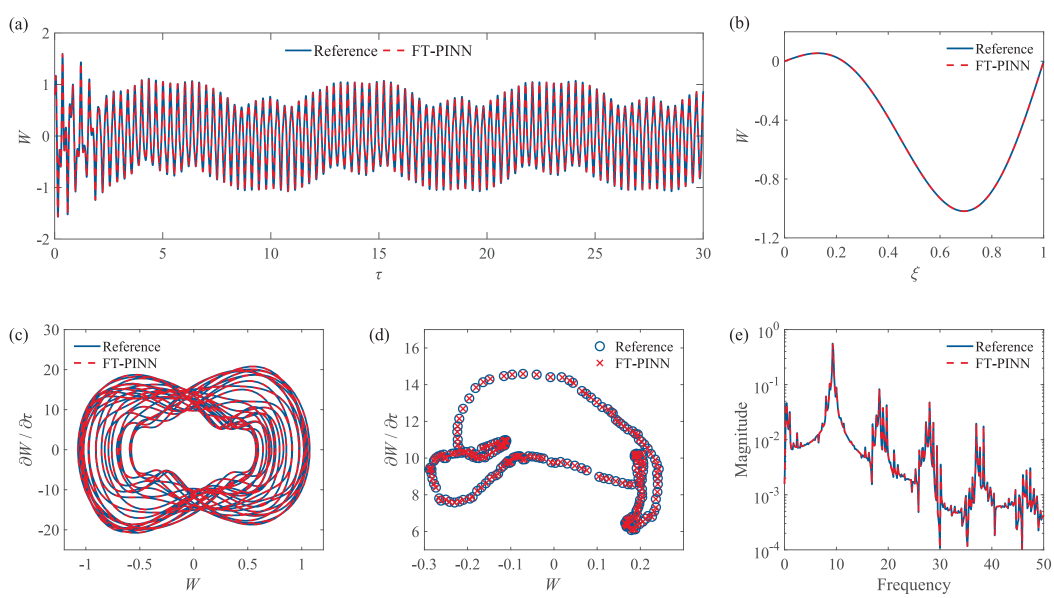

Fig. 10. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis under loading Case 5: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 20$; (c) phase plot; (d) Poincaré map; and (e) frequency spectrum.

4.3. Aerothermoelastic analysis with viscoelastic damping

The above examples use viscous damping as the damping model for the panel. In this section, we replace the viscous damping effect by the Kelvin–Voigt viscoelastic damping model [57] to verify the applicability of FT-PINN.

Using the Kelvin–Voigt viscoelastic damping model, the governing equation (1) of the panel is modified as

$$\begin{cases} \rho h \dfrac{\partial^2 w(x,t)}{\partial t^2} + D\left(1 + g\dfrac{\partial}{\partial t}\right)\dfrac{\partial^4 w(x,t)}{\partial x^4} \\ + \left(1 + g\dfrac{\partial}{\partial t}\right)\left[(N_{\mathrm{T}} - N_{\mathrm{G}})\dfrac{\partial^2 w(x,t)}{\partial x^2}\right] + \Delta p(x,t) = 0 \\ (x,t) \in (0,l) \times (0,T] \end{cases} \tag{24}$$

where g is the viscoelastic damping coefficient.

Equation (24) can be further transformed to a dimensionless form as follows:

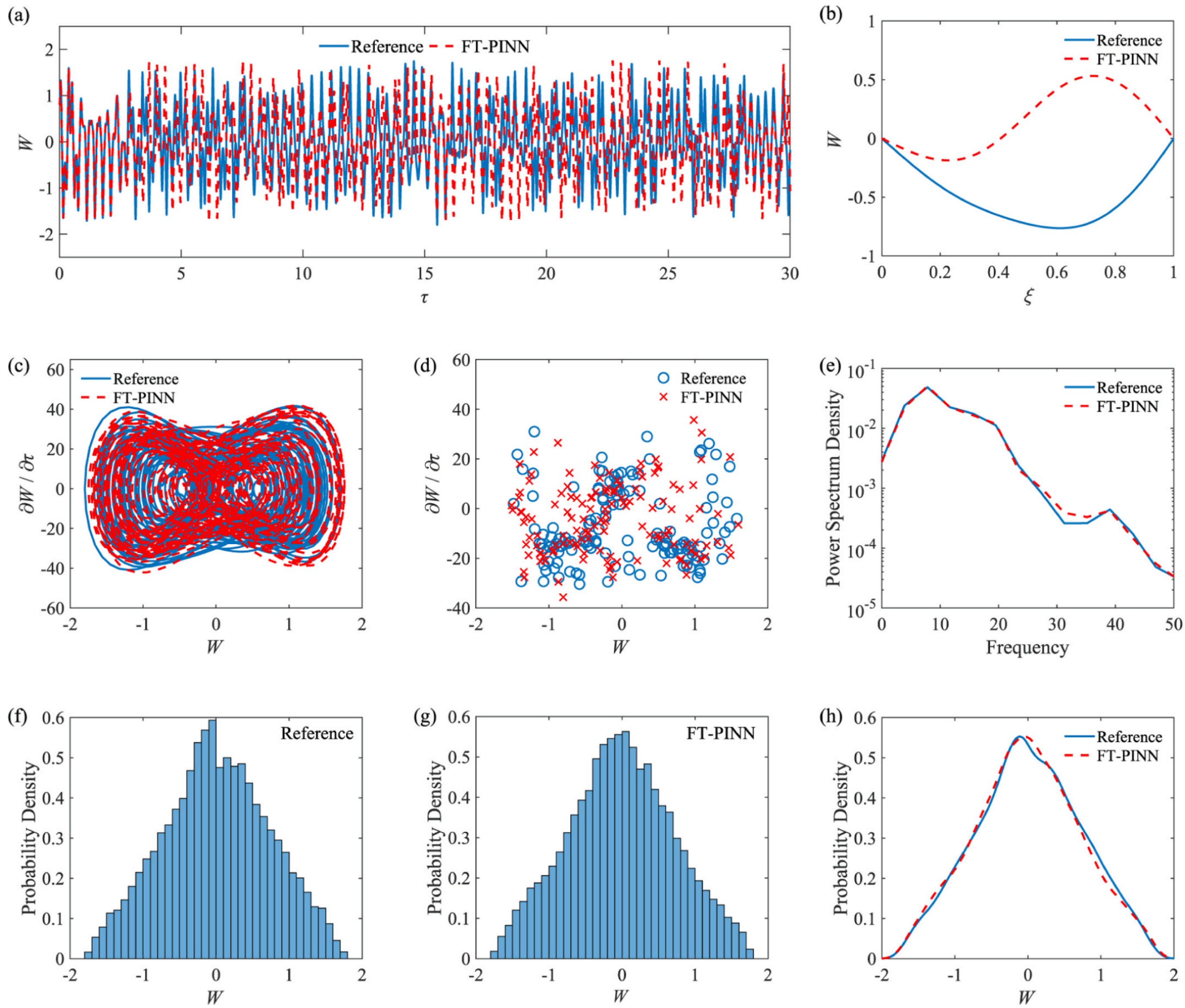

Fig. 11. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis under loading Case 6: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 20$; (c) phase plot; (d) Poincaré map; (e) power spectrum density; (f) probability density histogram of the reference solution; (g) probability density histogram of the FT-PINN solution; and (h) probability density curve obtained by kernel density estimation.

$$\begin{cases} \dfrac{\partial^2 W(\xi,\tau)}{\partial\tau^2} + \zeta\dfrac{\partial W(\xi,\tau)}{\partial\tau} + \left(1+G\dfrac{\partial}{\partial\tau}\right)\dfrac{\partial^4 W(\xi,\tau)}{\partial\xi^4} \\ +\left(1+G\dfrac{\partial}{\partial\tau}\right)\left\{\left[R_{\mathrm{T}} - 6\displaystyle\int_0^1 \left(\dfrac{\partial W(\xi,\tau)}{\partial\xi}\right)^2 \mathrm{d}\xi\right]\dfrac{\partial^2 W(\xi,\tau)}{\partial\xi^2}\right\} \\ +\lambda\dfrac{\partial W(\xi,\tau)}{\partial\xi} = 0 \\ (\xi,\tau)\in(0,1)\times(0,\Gamma] \end{cases} \tag{25}$$

where $G = g\sqrt{D/\rho h l^4}$ is the dimensionless viscoelastic damping coefficient. When solving Eq. (25) by the traditional Galerkin method, its discrete form needs to be derived. When solving Eq. (25) by the present FT-PINN method, we just need to modify the first PDE loss function in Eq. (17) as

$$\begin{cases} L_{p1}(\boldsymbol{\theta}) = \dfrac{1}{N_p}\sum_{k=1}^{N_p}\left\{\dfrac{\partial^2 \widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\Delta\Gamma^2\partial\widehat{\tau}^2} + \zeta\dfrac{\partial\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\Delta\Gamma\partial\widehat{\tau}} \right. \\ +\left(1+\dfrac{G\partial}{\Delta\Gamma\partial\widehat{\tau}}\right)\dfrac{\partial^4\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi^4} \\ +\left(1+\dfrac{G\partial}{\Delta\Gamma\partial\widehat{\tau}}\right)\left[\left(R_{\mathrm{T}} - 6\Phi_n(1,\widehat{\tau}_k^p) + 6\Phi_n(0,\widehat{\tau}_k^p)\right)\dfrac{\partial^2\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi^2}\right] \\ \left. +\lambda\dfrac{\partial\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi}\right\}^2 \\ (\xi_k^p,\widehat{\tau}_k^p)\in(0,1)\times(0,1] \end{cases} \tag{26}$$

In this example, the parameters of the panel, the network architecture and hyperparameters of FT-PINN are the same as those used in

Section 4.1. The viscoelastic damping coefficient is given by $g = $ 1E-04 s, so the value of G is equal to 3.07E-04. The thermal and aerodynamic loads are selected as Case 5 listed in Table 3. Fig. 12 compares the solutions predicted by FT-PINN with the reference solutions. It can be seen that, after replacing the viscous damping effect by the viscoelastic damping effect, the aerothermoelastic response changes from a quasi-periodic motion to a single-period LCO, and its frequency spectrum mainly contains a fundamental frequency of 8.4 along with its 3rd and 5th harmonics. The FT-PINN results demonstrate remarkable agreement with the reference solutions with a relative L_2 error of 3.63E-03, indicating the applicability of FT-PINN for aerothermoelastic analysis with various damping models.

4.4. Aerothermoelastic analysis with non-uniform thickness

The panel studied in the above examples has a uniform thickness. In this section, we consider a more complex panel with a non-uniform thickness. Consider a panel with linearly varying thickness as given by

$$h(x) = h_0\left(1+\beta\frac{x}{l}\right) \tag{27}$$

where h_0 is the thickness of the panel on the left end and β is the thickness varying factor. The governing equation of this panel is expressed as

$$\begin{cases} \rho h(x)\dfrac{\partial^2 w(x,t)}{\partial t^2} + c\dfrac{\partial w(x,t)}{\partial t} + \dfrac{\partial^2}{\partial x^2}\left[D(x)\dfrac{\partial^2 w(x,t)}{\partial x^2}\right] \\ +\dfrac{\beta}{\ln(1+\beta)}\left[\dfrac{2N_{\mathrm{T}}}{2+\beta} - \dfrac{Eh_0}{2l(1-\nu^2)}\displaystyle\int_0^l\left(\dfrac{\partial w(x,t)}{\partial x}\right)^2\mathrm{d}x\right]\dfrac{\partial^2 w(x,t)}{\partial x^2} \\ +\Delta p(x,t) = 0 \\ (x,t)\in(0,l)\times(0,T] \end{cases} \tag{28}$$

Equation (28) can be further transformed to a dimensionless form as follows:

$$\begin{cases} (1+\beta\xi)\dfrac{\partial^2 W(\xi,\tau)}{\partial \tau^2} + (C+\zeta)\dfrac{\partial W(\xi,\tau)}{\partial \tau} + (1+\beta\xi)^3\dfrac{\partial^4 W(\xi,\tau)}{\partial \xi^4} \\ +6\beta(1+\beta\xi)^2\dfrac{\partial^3 W(\xi,\tau)}{\partial \xi^3} + 6\beta^2(1+\beta\xi)\dfrac{\partial^2 W(\xi,\tau)}{\partial \xi^2} \\ +\dfrac{\beta}{\ln(1+\beta)}\left[\dfrac{2R_{\mathrm{T}}}{2+\beta} - 6\displaystyle\int_0^1\left(\dfrac{\partial W(\xi,\tau)}{\partial \xi}\right)^2\mathrm{d}\xi\right]\dfrac{\partial^2 W(\xi,\tau)}{\partial \xi^2} \\ +\lambda\dfrac{\partial W(\xi,\tau)}{\partial \xi} = 0 \\ (\xi,\tau)\in(0,1)\times(0,\Gamma] \end{cases} \tag{29}$$

Note that the dimensionless variables in Eq. (29) are not expressed as the same as those in Eq. (4). They are expressed by replacing h and D with h_0 and $D_0 = Eh_0^3/12(1-\nu^2)$ in the dimensionless variables of Eq. (4). The solution of Eq. (29) using conventional numerical techniques, such as the finite element method, imposes stringent demands on spatial discretization, necessitating a high density of elements to capture the non-uniform thickness. Similarly, applying the Galerkin method or the Rayleigh – Ritz method to this equation requires a complex and cumbersome discretization process. In contrast, the FT-PINN method can easily address this problem by simply modifying the first PDE loss function in Eq. (17) as

$$\begin{cases} L_{p1}(\boldsymbol{\theta}) = \dfrac{1}{N_p}\displaystyle\sum_{k=1}^{N_p}\left\{\dfrac{(1+\beta\xi_k^p)\partial^2\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\Delta\Gamma^2\partial\widehat{\tau}^2}\right. \\ \quad + (C+\zeta)\dfrac{\partial\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\Delta\Gamma\partial\widehat{\tau}} + (1+\beta\xi_k^p)^3\dfrac{\partial^4\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi^4} \\ \quad + 6\beta(1+\beta\xi_k^p)^2\dfrac{\partial^3\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi^3} + 6\beta^2(1+\beta\xi_k^p)\dfrac{\partial^2\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi^2} \\ \quad + \dfrac{\beta}{\ln(1+\beta)}\left[\dfrac{2R_{\mathrm{T}}}{2+\beta} - 6\Phi_n(1,\widehat{\tau}_k^p) + 6\Phi_n(0,\widehat{\tau}_k^p)\right]\dfrac{\partial^2\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi^2} \\ \quad \left. + \lambda\dfrac{\partial\widehat{W}_n(\xi_k^p,\widehat{\tau}_k^p)}{\partial\xi}\right\}^2 \\ (\xi_k^p,\widehat{\tau}_k^p)\in(0,1)\times(0,1] \end{cases} \tag{30}$$

In this example, the parameters of the panel are same as those used in previous examples except the thickness. Here, the thickness linearly varies from 1.8 mm to 2.2 mm with the same average thickness as previous examples, i.e., $h_0 = 1.8$ mm and $\beta = 2/9$. The network architecture and hyperparameters of FT-PINN are the same as those used in Section 4.1. The aerothermoelastic response of this panel is solved by FT-PINN under the loading Case 5, and the results are compared to the panel with a uniform thickness of 2 mm. Because the dimensionless variables of the panels with uniform and non-uniform thicknesses have different expressions, we transform their aerothermoelastic responses to dimensional forms for comparison. As illustrated in Fig. 13, although the two panels have the same average thickness, their responses are much different. Unlike the quasi-periodic motion of the uniform panel, the response of the non-uniform panel is an off-center single-period LCO, and its frequency spectrum mainly contains the first five harmonics. We calculate the peak-peak values of the vibration displacements over the two panels and compare them in Fig. 13(b). It can be found that the deformation of the non-uniform panel is significantly lower than the uniform panel. This example demonstrates the potential of the FT-PINN method for aerothermoelastic analysis in complex scenarios, as well as its applicability in structural design for response suppression.

4.5. Aerothermoelastic analysis with time-varying parameters

The structural and loading parameters in the above examples are constant. It means these aerothermoelastic systems are time-invariant. In this section, we consider a panel under time-varying loading conditions and investigate the validity of FT-PINN for solving the time-varying system. Suppose that the thermally induced in-plane force N_{T} and the Mach number Ma of airflow are time-varying, so the speed of airflow V is also time-varying, and the governing equation (1) of the panel needs to be rewritten as

$$\begin{cases} \rho h\dfrac{\partial^2 w(x,t)}{\partial t^2} + c\dfrac{\partial w(x,t)}{\partial t} + D\dfrac{\partial^4 w(x,t)}{\partial x^4} + [N_{\mathrm{T}}(t) - N_{\mathrm{G}}]\dfrac{\partial^2 w(x,t)}{\partial x^2} \\ +\dfrac{\rho_{\mathrm{a}}V(t)^2}{\sqrt{Ma(t)^2-1}}\left\{\dfrac{\partial w}{\partial x} + \dfrac{Ma(t)^2-2}{\left[Ma(t)^2-1\right]V(t)}\dfrac{\partial w}{\partial t}\right\} = 0 \\ (x,t)\in(0,l)\times(0,T] \end{cases} \tag{31}$$

Equation (31) can be further transformed to a dimensionless form as follows:

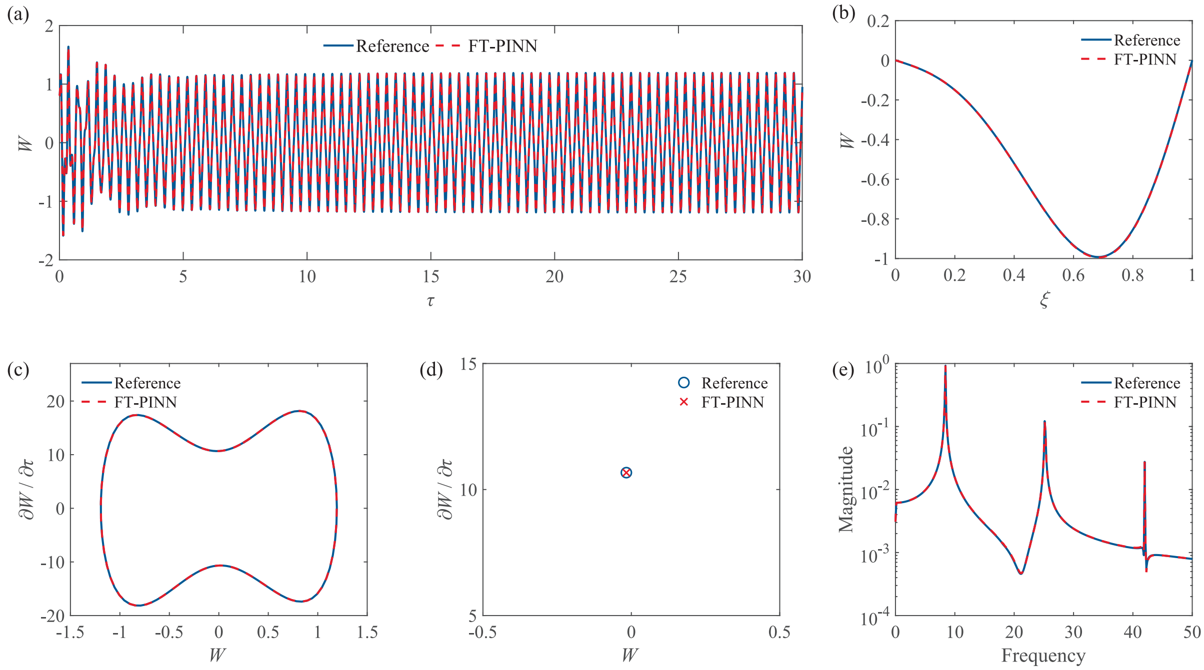

Fig. 12. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis with viscoelastic damping under loading Case 5: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 20$; (c) phase plot; (d) Poincaré map; and (e) frequency spectrum.

$$\begin{cases} \dfrac{\partial^2 W(\xi,\tau)}{\partial\tau^2} + [C+\zeta(\tau)]\dfrac{\partial W(\xi,\tau)}{\partial\tau} + \dfrac{\partial^4 W(\xi,\tau)}{\partial\xi^4} \\ + \left[R_{\mathrm{T}}(\tau) - 6\displaystyle\int_0^1 \left(\dfrac{\partial W(\xi,\tau)}{\partial\xi}\right)^2 \mathrm{d}\xi\right]\dfrac{\partial^2 W(\xi,\tau)}{\partial\xi^2} + \lambda(\tau)\dfrac{\partial W(\xi,\tau)}{\partial\xi} = 0 \\ (\xi,\tau)\in(0,1)\times(0,\Gamma] \end{cases} \tag{32}$$

where $\lambda(\tau)$ and $\zeta(\tau)$ are expressed as

$$\begin{cases} \lambda(\tau) = \dfrac{\rho_{\mathrm{a}} V(t)^2 l^3}{D\sqrt{Ma(t)^2 - 1}} \\ \zeta(\tau) = \dfrac{\rho_{\mathrm{a}} V(t)\left[Ma(t)^2 - 2\right]}{\left[Ma(t)^2 - 1\right]^{3/2}}\sqrt{\dfrac{l^4}{D\rho h}} \end{cases} \tag{33}$$

in which $t = \tau\sqrt{\rho h l^4 / D}$.

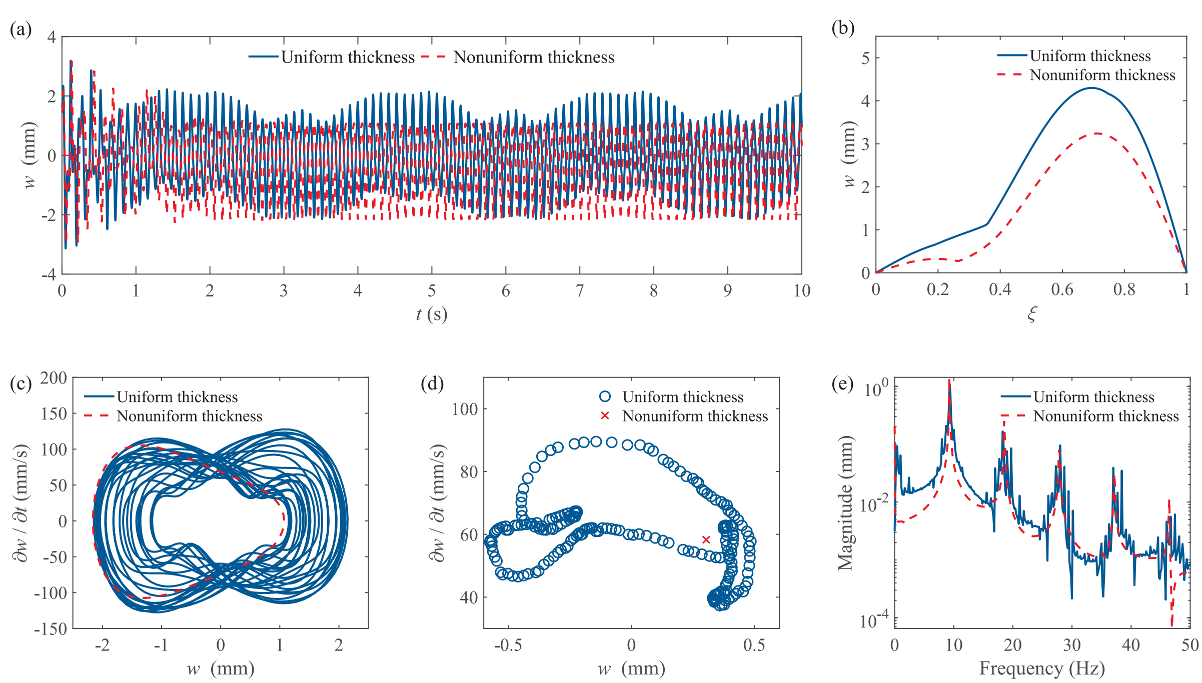

Fig. 13. Comparison of aerothermoelastic responses predicted by FT-PINN for panels with uniform and non-uniform thicknesses under loading Case 5: (a) time history of displacement at $\xi = 0.7$; (b) peak-peak value of the vibration displacement over the panel; (c) phase plot; (d) Poincaré map; and (e) frequency spectrum.

Accordingly, the first PDE loss function in Eq. (17) needs to be modified as

$$\begin{cases} L_{p1}(\boldsymbol{\theta}) = \dfrac{1}{N_p}\sum_{k=1}^{N_p}\left\{\dfrac{\partial^2 \widehat{W}_n(\xi_k^p, \widehat{\tau}_k^p)}{\Delta\Gamma^2\partial\widehat{\tau}^2}\right. \\ \quad + \left[C + \zeta(\widehat{\tau}_k^p\Delta\Gamma + \Gamma_{n-1})\right]\dfrac{\partial\widehat{W}_n(\xi_k^p, \widehat{\tau}_k^p)}{\Delta\Gamma\partial\widehat{\tau}} + \dfrac{\partial^4\widehat{W}_n(\xi_k^p, \widehat{\tau}_k^p)}{\partial\xi^4} \\ \quad + \left[R_{\mathrm{T}}(\widehat{\tau}_k^p\Delta\Gamma + \Gamma_{n-1}) - 6\Phi_n(1, \widehat{\tau}_k^p) + 6\Phi_n(0, \widehat{\tau}_k^p)\right]\dfrac{\partial^2\widehat{W}_n(\xi_k^p, \widehat{\tau}_k^p)}{\partial\xi^2} \\ \quad \left. + \lambda(\widehat{\tau}_k^p\Delta\Gamma + \Gamma_{n-1})\dfrac{\partial\widehat{W}_n(\xi_k^p, \widehat{\tau}_k^p)}{\partial\xi}\right\}^2 \\ (\xi_k^p, \widehat{\tau}_k^p) \in (0,1)\times(0,1] \end{cases} \tag{34}$$

As can be seen from Eq. (34), to solve Eq. (32) with time-varying parameters by the present FT-PINN method, we just need to get the values of these parameters at each sample point and substitute them into the PDE loss function.

In this example, the structural parameters of the panel and the network architecture of FT-PINN are the same as those used in Section 4.1. The density and sound speed of airflow are given by $\rho_a = 0.089\ \mathrm{kg/m^3}$ and $a_\infty = 295\ \mathrm{m/s}$. As time evolves, we make N_{T} linearly increase from 1000 N/m to 3000 N/m and Ma linearly increase from 2 to 3. The value ranges of the corresponding dimensionless parameters for thermal and aerodynamic loads are $R_{\mathrm{T}} \in [12.1, 36.2]$, $\lambda \in [215.7, 297.2]$ and $\zeta \in [0.748, 0.902]$. Fig. 14 compares the aerothermoelastic responses predicted by FT-PINN with the reference solutions. As illustrated in Fig. 14(a), the panel is stable for $0 \le \tau \le 5$ and the aerothermoelastic response gradually decays because the thermal and aerodynamic loads are low. After $\tau = 5$, the panel becomes dynamically unstable and vibrates with increasing amplitude as the thermal and aerodynamic loads increase. The evolution of the aerothermoelastic response can also be observed in Fig. 14(c) and (d). Fig. 14 shows the excellent agreement between the FT-PINN results with the reference one, and the relative L_2 error is only 1.67E-03. It proves that the proposed FT-PINN method is applicable to the aerothermoelastic analysis of time-varying systems.

5. Conclusion

A novel FT-PINN method is proposed herein for nonlinear and multi-frequency aerothermoelastic analysis. The FT-PINN framework transforms the governing equations of aerothermoelastic problems from an integro-differential form into a system of PDEs by introducing a primitive function. To ensure solution accuracy over long durations, a time-sequential training strategy is employed, further enhanced by the normalization and hard-constraint techniques. In addition, the method incorporates random Fourier feature mapping to pre-process the input spatiotemporal coordinates, significantly improving its ability to capture the nonlinear and multi-frequency characteristics inherent in aerothermoelastic responses.

We conducted a series of numerical experiments to validate the effectiveness and performance of the FT-PINN method for nonlinear aerothermoelastic analysis. The key advantage of FT-PINN is its ability to provide accurate aerothermoelastic responses with nonlinear and multi-frequency properties over long-duration periods. Compared to existing PINN-based methods, FT-PINN achieves two to three orders of magnitude reduction in relative L_2 error. Furthermore, the method effectively solves aerothermoelastic problems under diverse loading conditions, accurately predicting responses such as decay to zero, thermal buckling, LCOs, quasi-periodic and chaotic motions. Complex aerothermoelastic problems involving viscoelastic damping, non-uniform thickness and time-varying parameters have also been successfully addressed using FT-PINN. These results demonstrate the excellent accuracy, effectiveness, and applicability of the proposed method.

Despite these advancements, technical limitations remain in the present FT-PINN method. For example, the hard-constraint technique employed in FT-PINN is based on trigonometric functions and only suitable for structures with regular geometric shapes. Future work will

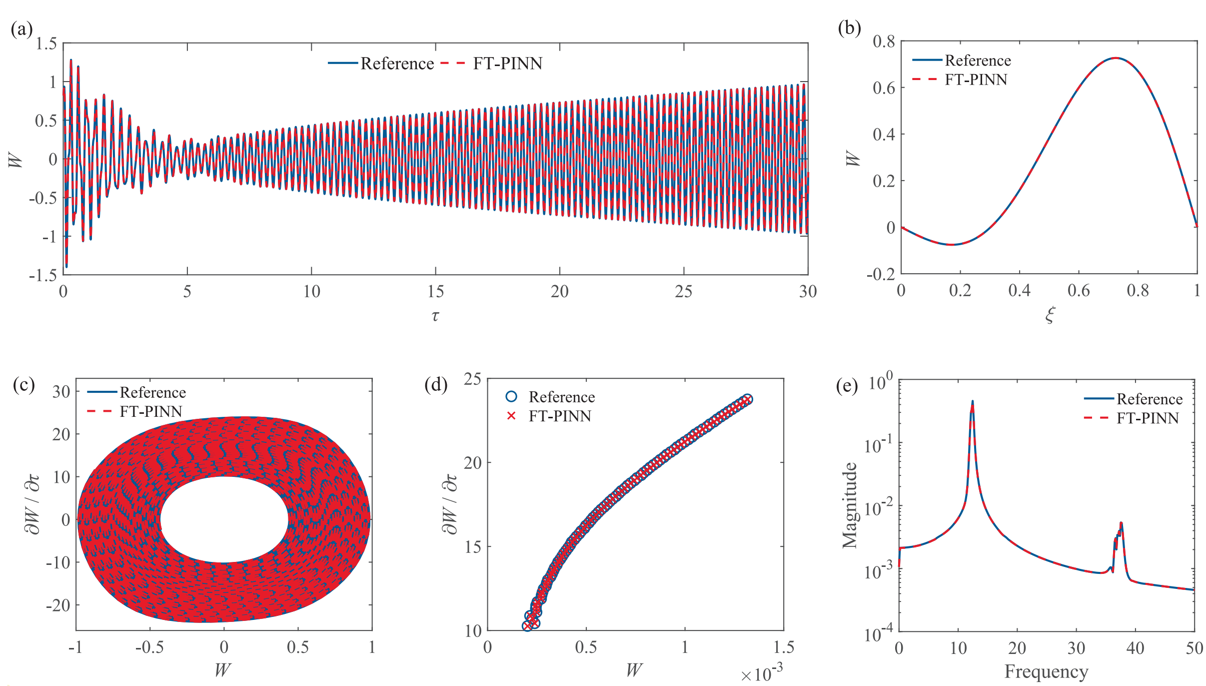

Fig. 14. Comparison of reference and FT-PINN predicted results for aerothermoelastic analysis with time-varying parameters: (a) time history of displacement at $\xi = 0.7$; (b) instantaneous deformation shape at $\tau = 20$; (c) phase plot; (d) Poincaré map; and (e) frequency spectrum. (Note: Curves in figures (c), (d) and (e) are computed by using the response at $\xi = 0.7$ after $\tau = 10$.).

extend the present method by incorporating universal approximate distance functions [70,71] to constrain boundary conditions, thus enabling its application to complex-shaped structures. This approach also struggles to solve strongly nonlinear dynamic problems with localized high-gradient solutions. This limitation can be alleviated by improving its sampling strategy, for instance, by adding more sample points around high-gradient regions, as demonstrated in methods like GW-PINN [29], PIPINN [72], and PMTPINN [73]. Furthermore, this method does not yet match the accuracy and efficiency of well-established numerical methods, e.g., the Galerkin and finite element methods. However, as an extension of the PINN framework, the FT-PINN approach exhibits its inherent advantages and offers a meshless and universal approach, eliminating the cumbersome processes of PDE discretization and fine mesh generation required by traditional means. With continued advances in artificial intelligence, FT-PINN is expected to become a promising tool for tackling complex nonlinear dynamic problems in a variety of structures subjected to multi-physics coupling loads.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Zhaolin Chen: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Zhicheng Yang:** Writing – review & editing, Methodology, Investigation. **Yiting Zhang:** Writing – review & editing, Methodology, Investigation. **Changning Liu:** Writing – review & editing, Methodology, Investigation. **Zhichun Yang:** Writing – review & editing, Methodology, Investigation. **Siu-Kai Lai:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Algorithm of FT-PINN

Algorithm A1: FT-PINN algorithm.

Inputs: Parameters of dimensionless PDEs, loss function weights, number of sample points, simulation duration Γ, time interval $\Delta\Gamma$, mapping dimension m and standard deviations (σ_{ξ} and $\sigma_{\hat{\tau}}$) of the random Fourier feature mapping technique, number of hidden layers, number of neurons in each hidden layer, initial displacement $h_1(\xi)$ and initial velocity $h_2(\xi)$.
Outputs: Trained neural network parameters and the solution of PDEs.
1: Build and initialize a fully-connected neural network $\Psi(\boldsymbol{x}, \boldsymbol{\theta})$.
2: Calculate the number of time segments by $M = \Gamma/\Delta\Gamma$.
3: Collect sample points $(\xi_k^p, \hat{\tau}_k^p)$, $(\xi_k^b, \hat{\tau}_k^b)$ and ξ_k^i in $(\xi, \hat{\tau}) \in [0,1] \times [0,1]$.
4: Collect mapping factors $\boldsymbol{k}$ and $\boldsymbol{f}$ from Gaussian distributions $\mathcal{N}\left(0, \sigma_{\xi}^2\right)$ and $\mathcal{N}\left(0, \sigma_{\hat{\tau}}^2\right)$.
5: Set initial conditions for the first time segment by
$\widehat{W}_0(\xi_k^i, 1) = h_1(\xi_k^i)$, $\partial\widehat{W}_0(\xi_k^i, 1)/\partial\hat{\tau} = h_2(\xi_k^i)\Delta\Gamma$.
6: **For** $n = 1, 2, \cdots, M$ **do**
(a) Compute the Fourier feature mapping values $\boldsymbol{a}$ and $\boldsymbol{b}$ of the coordinates of the sample points by Eq. (20);
(b) Input $\boldsymbol{a}$ to the neural network Ψ and get the outputs W_{ξ} and Φ_{ξ};
(c) Input $\boldsymbol{b}$ to the neural network Ψ and get the outputs $W_{\hat{\tau}}$ and $\Phi_{\hat{\tau}}$;
(d) Compute W_n and Φ_n by $W_n = W_{\xi}W_{\hat{\tau}}$ and $\Phi_n = \Phi_{\xi}\Phi_{\hat{\tau}}$, respectively;
(e) Impose hard constraints to W_n and get $\widehat{W}_n$ by Eq. (15);
(f) Compute the total loss by substituting $\widehat{W}_n$ and Φ_n into Eqs. (16)–(19);
(g) Update the trainable parameters $\boldsymbol{\theta}$ of the network until the optimizer meets its convergence criteria;
(h) Input $(\xi_k^b, 1)$ to the trained neural network and get $\widehat{W}_n(\xi_k^i, 1)$ and $\partial\widehat{W}_n(\xi_k^i, 1)/\partial\hat{\tau}$;
(i) Save the trained neural network for the n^{th} time segment;
End.
7: Predict the solution of PDEs in the entire spatiotemporal domain by using the trained neural networks.

Appendix B. Selection of hyperparameters in FT-PINN

As shown in Sections 3 and 4, the proposed FT-PINN method contains many hyperparameters, such as the time interval $\Delta\Gamma$, the loss function weights (λ_{p2}, λ_b, λ_{i1} and λ_{i2}), the mapping dimension m, and the standard deviations (σ_ξ and $\sigma_{\tilde{\tau}}$) of the random Fourier feature mapping technique. These hyperparameters significantly influence the solution accuracy of FT-PINN. We select the proper values of these hyperparameters through a trial-and-error process and the control variate method. In this section, we take Case 4 (discussed in Section 4.1) as an example and present representative results to illustrate parametric sensitivity and identify the optimal hyperparameters.

Time interval

The time interval $\Delta\Gamma$ is the key parameter in the time-sequential training strategy of FT-PINN. Fig. B1 shows the relative L_2 errors of FT-PINN for $\tau \in [0,3]$ with different time intervals of $\Delta\Gamma = 0.075$, 0.15, 0.3 and 0.6. It can be seen that, as $\Delta\Gamma$ decreases from 0.6 to 0.15, the relative L_2 error gradually decreases. Generally, shortening the time interval is benefit to reduce the loss function and enhance the solution accuracy. However, a smaller $\Delta\Gamma$ implies that more time segments are required, raising the cumulative error of the initial conditions in each time segment. As illustrated in Fig. B1, when $\Delta\Gamma$ decreases from 0.15 to 0.075, the relative L_2 error slightly increases. Therefore, a time interval of $\Delta\Gamma = 0.15$ is selected to guarantee the best solution accuracy.

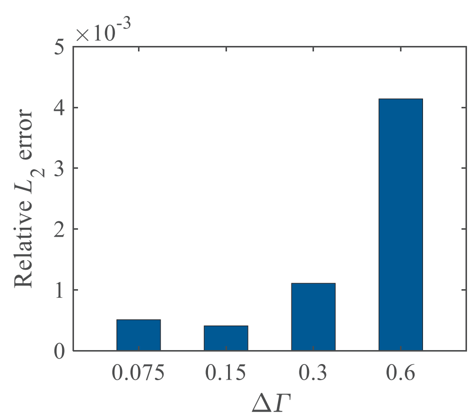

Fig. B1. Relative L_2 error of FT-PINN for $\tau \in [0,3]$ with different time intervals $\Delta\Gamma$.

Hyperparameters of random Fourier feature mapping

The mapping dimension m and standard deviations (σ_{ξ} and $\sigma_{\hat{\tau}}$) are the core parameters of the random Fourier feature mapping. We choose the first time segment of Case 4 as an example to investigate the solution accuracy with different values of m, σ_{ξ} and $\sigma_{\hat{\tau}}$. As shown in Fig. B2(a), the relative L_2 error gradually decreases with the increase of m. When m is larger than 128, the result shows no obvious improvement, so $m = 128$ is selected considering that a larger m leads to more computational cost. As shown in Fig. B2(b) and (c), the relative L_2 error firstly decreases and then increases as σ_{ξ} and $\sigma_{\hat{\tau}}$ increase from 0.25 to 8. Compared to $\sigma_{\hat{\tau}}$, the relative L_2 error is more sensitive to σ_{ξ}. When σ_{ξ} increases from 1 to 8, the relative L_2 error rapidly increases by three orders of magnitude. The best solution accuracy is achieved with $\sigma_{\xi} = 1$ and $\sigma_{\hat{\tau}} = 2$, which tune the wavenumber and frequency bands of the random Fourier feature mapping technique to proper ranges. Hence, the standard deviations are set to $\sigma_{\xi} = 1$ and $\sigma_{\hat{\tau}} = 2$.

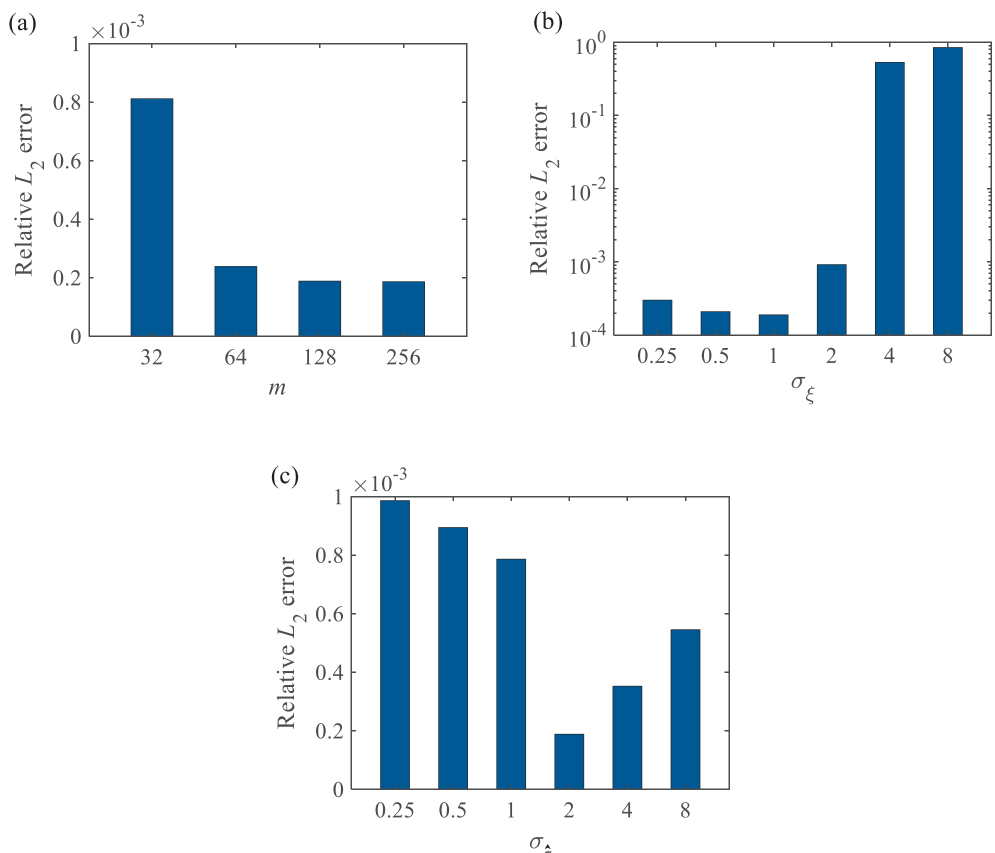

Fig. B2. Relative L_2 error of FT-PINN in the first time segment of Case 4 with different hyperparameters of the random Fourier feature mapping: (a) mapping dimension m; (b) standard deviation σ_{ξ}; and (c) standard deviation $\sigma_{\hat{\tau}}$. In each figure, except the parameter studied, other parameters are given by $m = 128$, $\sigma_{\xi} = 1$ and $\sigma_{\hat{\tau}} = 2$.

Loss function weights

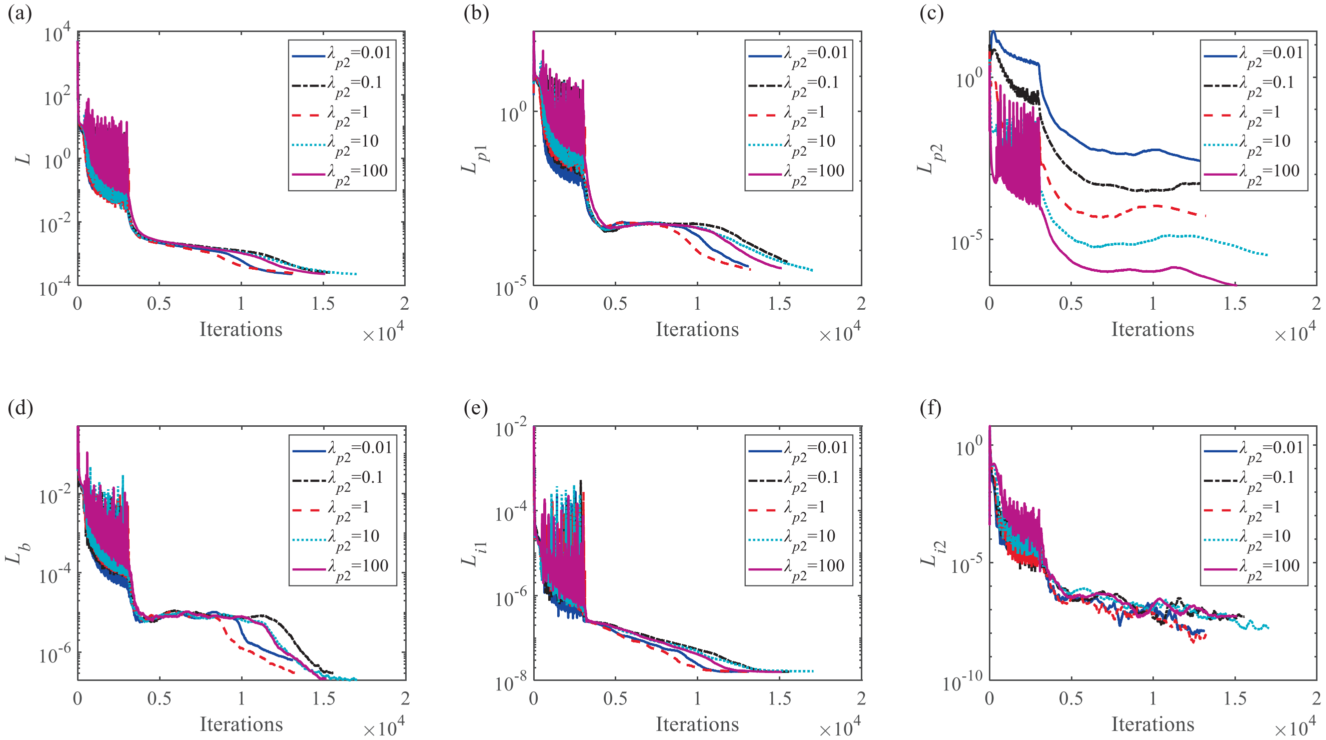

Fig. B3. Convergence plots of the total loss function and its components for FT-PINN in the first time segment of Case 4 with varying λ_{p2}: (a) total loss L; (b) first PDE loss L_{p1}; (c) second PDE loss L_{p2}; (d) boundary curvature loss L_b; (e) initial displacement loss L_{i1}; and (f) initial velocity loss L_{i2}. In each figure, except the weight λ_{p2}, other weights are given by $\lambda_b = 10$, $\lambda_{i1} = 10000$ and $\lambda_{i2} = 10$.

According to Eq. (19), the total loss function L of FT-PINN has five components, i.e., L_{p1}, L_{p2}, L_b, L_{i1} and L_{i2}. The weight of the first PDE loss L_{p1} is given by one. The weights of the other four components (i.e., λ_{p2}, λ_b, λ_{i1} and λ_{i2}) balance the effects of different physics laws, and influence the evolution of each loss and the solution accuracy. Taking λ_{p2}, the weight of the second PDE loss, as an example, Fig. B3 compares the convergence plots of the total loss function and its five components in the first time segment of Case 4 with $\lambda_{p2} = 0.01$, 0.1, 1, 10 and 100. It can be seen from Fig. B3(c), higher λ_{p2} enables FT-PINN to better learn the second PDE and leads to a reduction in L_{p2}. However, the evolution of other loss components and the total loss have no simple trend of change with the increasing λ_{p2}, indicating that it is hard to determine the loss function weights through these convergence plots. Therefore, we directly compare the solution accuracy with different weight values to find the optimal combination. Fig. B4 shows the relative L_2 errors of FT-PINN in the first time segment of Case 4 with different weight values. For each increasing loss function weight, the relative L_2 error firstly decreases and then increases. Compared to λ_{p2}, λ_b and λ_{i2}, the relative L_2 error is more sensitive to λ_{i1}. According to Fig. B4, the optimal loss function weights $\lambda_{p2} = 1$, $\lambda_b = 10$, $\lambda_{i1} = 10000$ and $\lambda_{i2} = 10$ are selected.

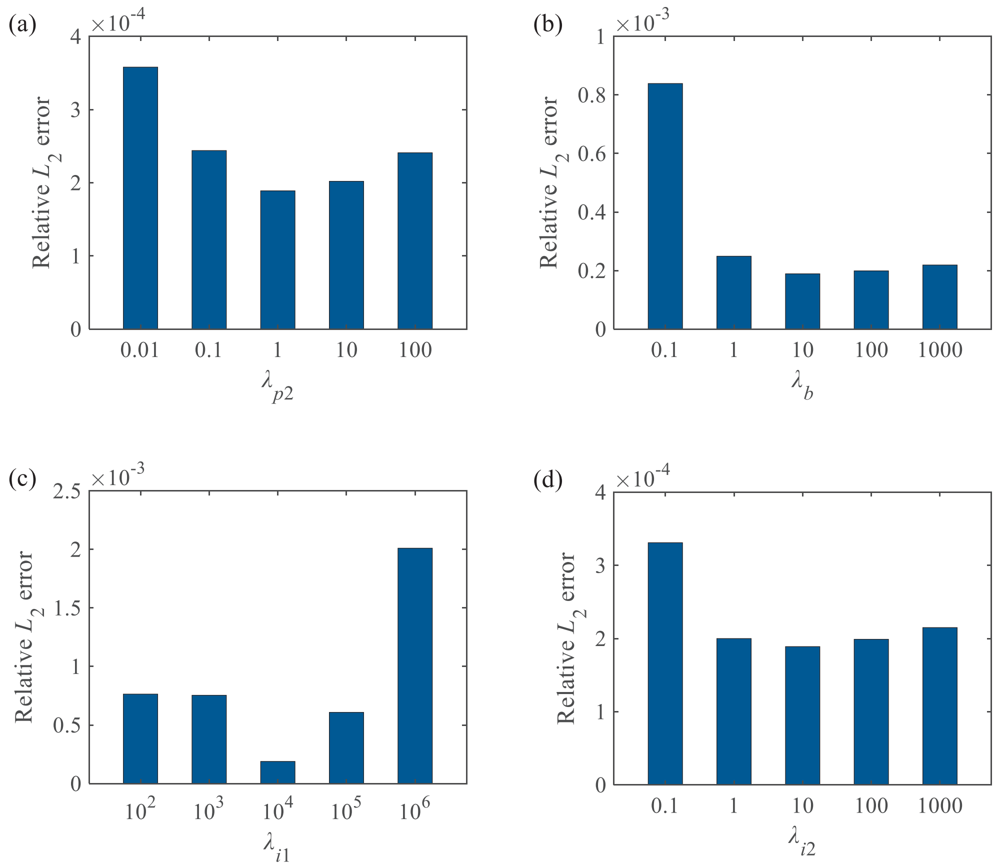

Fig. B4. Relative L_2 error of FT-PINN in the first time segment of Case 4 with different loss function weights: (a) λ_{p2}, weight of the second PDE loss; (b) λ_b, weight of the boundary curvature loss; (c) λ_{i1}, weight of the initial displacement loss; and (d) λ_{i2}, weight of the initial velocity loss. In each figure, except the weight studied, other weights are given by $\lambda_{p2} = 1$, $\lambda_b = 10$, $\lambda_{i1} = 10000$ and $\lambda_{i2} = 10$.

Appendix C. Results of aerothermoelastic analysis under various loading conditions

Figs. C1-C5 present representative solutions and error contours of FT-PINN for aerothermoelastic analysis under Cases 1, 2, 3, 5, and 6, as listed in Table 3. Note that the contours for Case 4 are provided in Section 4.1 and are not repeated here.

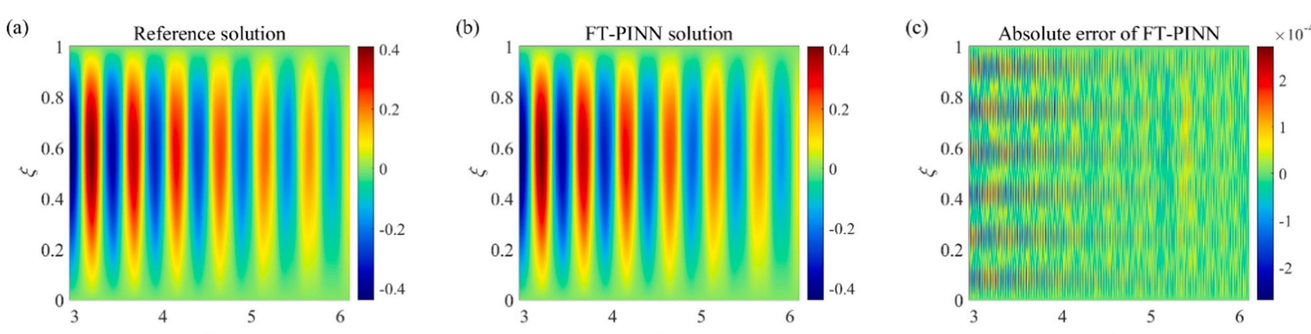

Fig. C1. Results of aerothermoelastic analysis under loading Case 1 for $\tau \in [3, 6]$: (a) reference solution; (b) FT-PINN solution; and (c) absolute error.

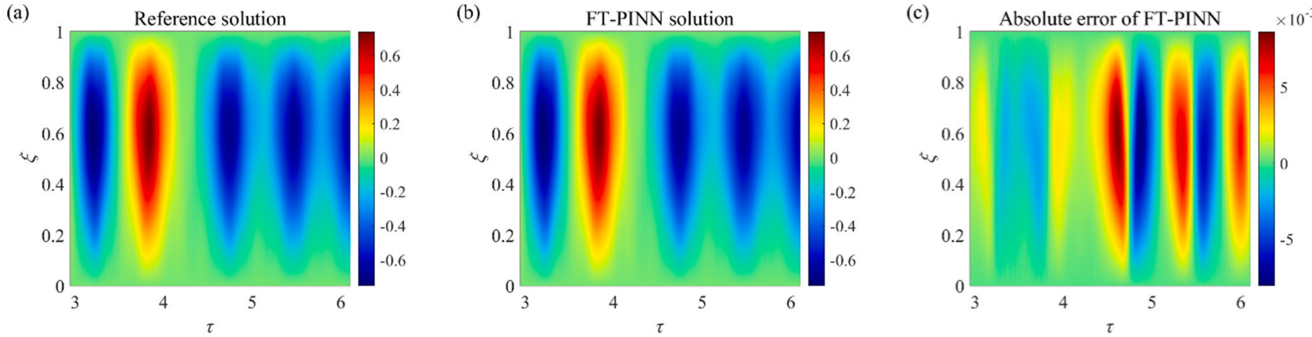

Fig. C2. Results of aerothermoelastic analysis under loading Case 2 for $\tau \in [3, 6]$: (a) reference solution; (b) FT-PINN solution; and (c) absolute error.

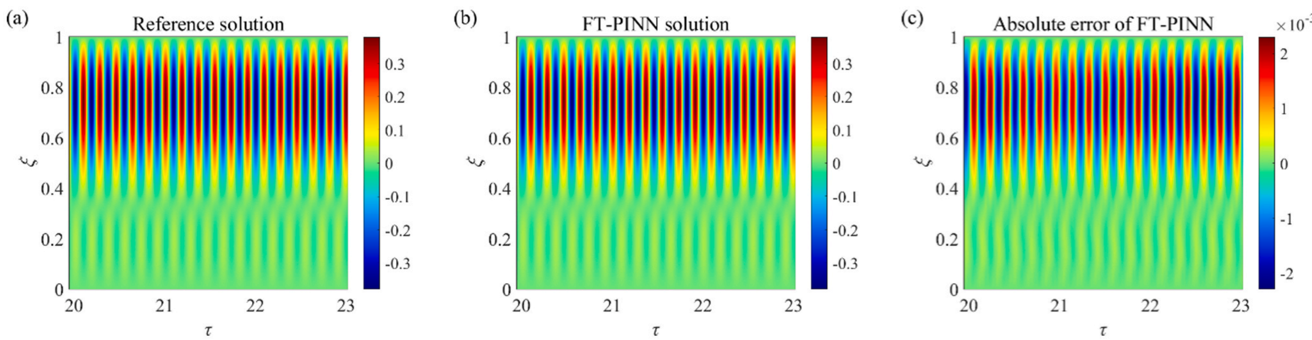

Fig. C3. Results of aerothermoelastic analysis under loading Case 3 for $\tau \in [20, 23]$: (a) reference solution; (b) FT-PINN solution; and (c) absolute error.

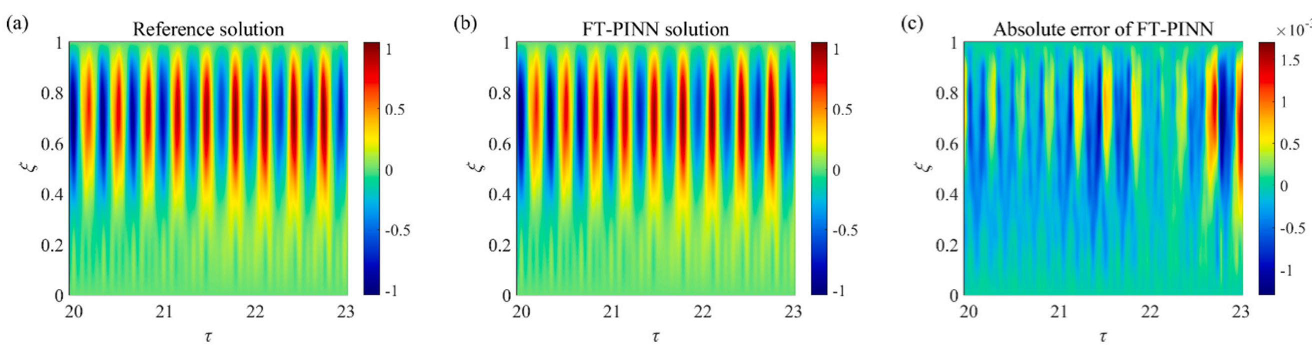

Fig. C4. Results of aerothermoelastic analysis under loading Case 5 for $\tau \in [20, 23]$: (a) reference solution; (b) FT-PINN solution; and (c) absolute error.

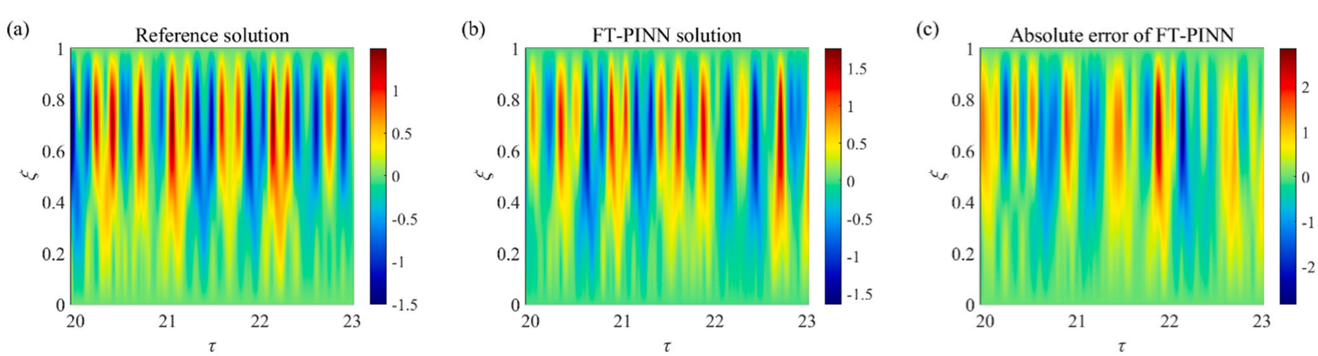

Fig. C5. Results of aerothermoelastic analysis under loading Case 6 for $\tau \in [20, 23]$: (a) reference solution; (b) FT-PINN solution; and (c) absolute error.

Data availability

Data will be made available on request.

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